CHAPTER 49
AUXILIARY POWER UNIT

CONTENTS

Page

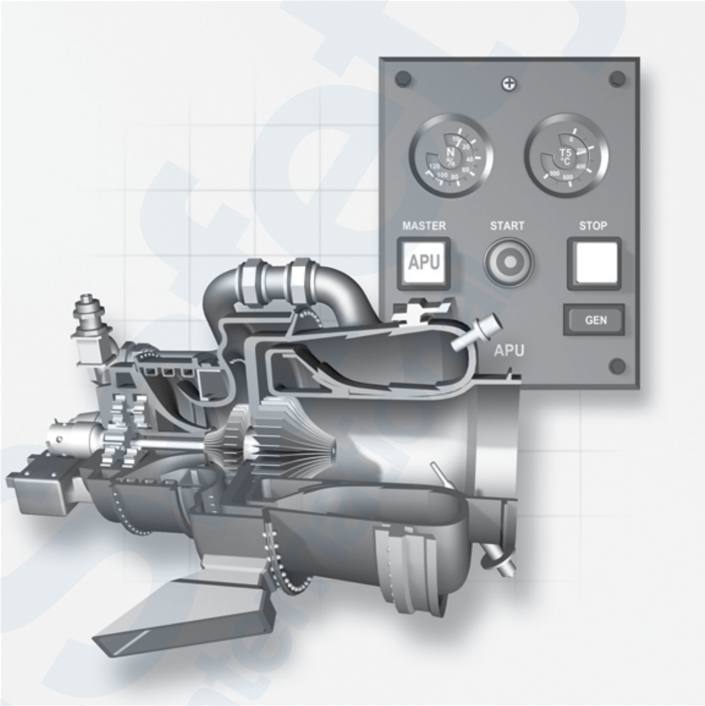
Page

Page

ILLUSTRATIONS

TABLES

CHAPTER 49
AUXILIARY POWER UNIT



49-00-00 INTRODUCTION

The APS 1000 Model (T62T-46C12) APU is a turbine-powered system used to supply DC electrical power for the aircraft electrical system and bleed air for the Environmental Control System (ECS). The APU is a non-essential system and can only be operated when the aircraft is on the ground.

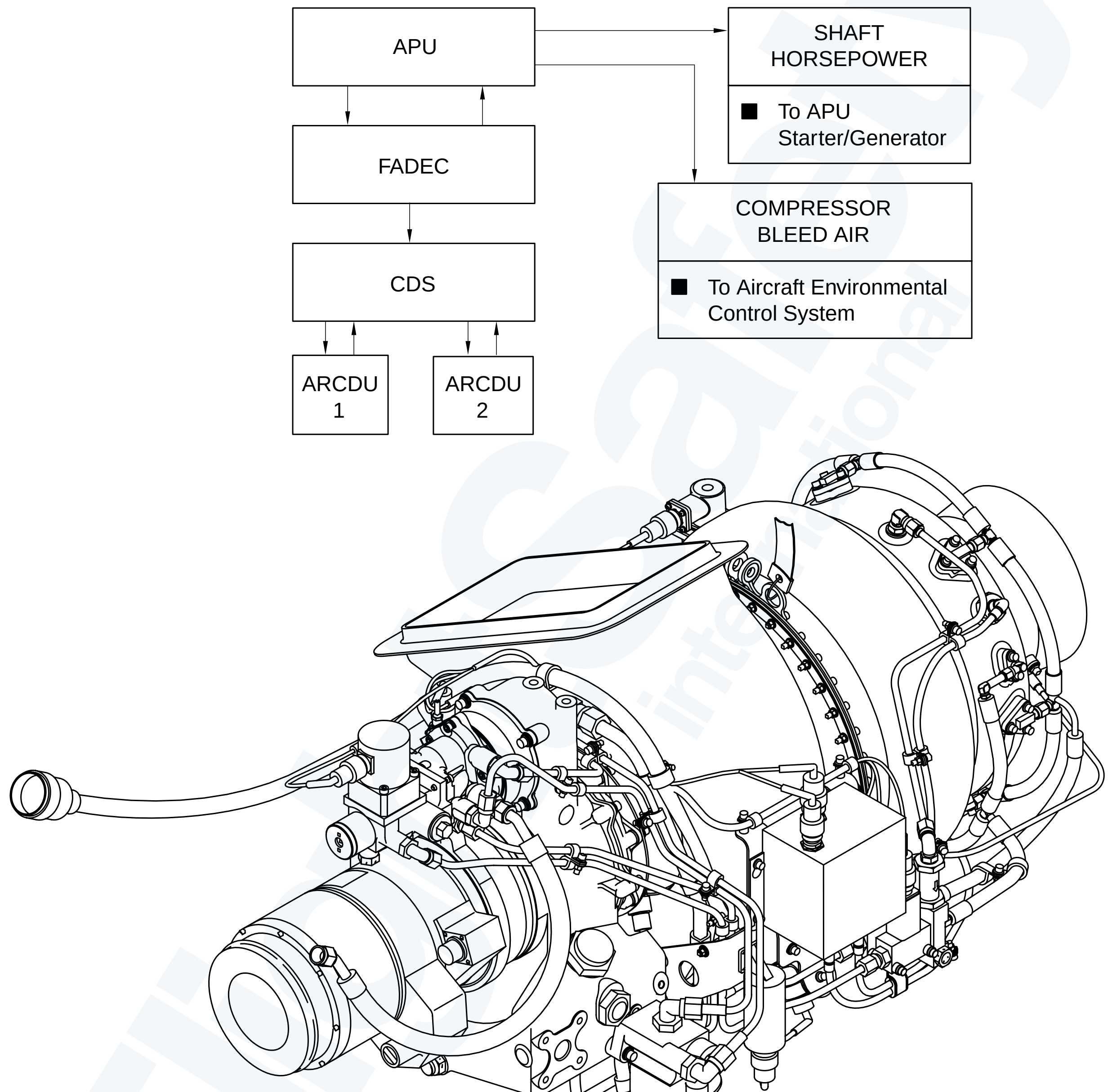


Figure 49-1. APU - Synoptic

GENERAL

Refer to:

- Figure 49-1. APU - Synoptic.
- Table 49-1. Leading Particulars.

The APU engine and its associated components are installed in the tail cone. The gas turbine engine drives the gearbox module. The gearbox module reduces the turbine RPM and drives a starter/generator, APU fuel pump and an internal oil pump. The APU gives bleed air to the ECS and 28 VDC to the right main feeder bus.

The APU has the components that follow:

- An air induction/compression system
- An exhaust system
- A reduction gearbox
- A lubrication system
- An ignition system
- A combination starter/generator
- Full Authority Digital Electronic Controller (FADEC)
- A control/monitoring system
- A control panel
- Interconnect harness
- Centralized Diagnostic System (CDS) interface.

Leading Particulars	
Weight	140 lbs (dry)
Operating Altitude	—1,000 to 15,000 ft (Ground Mode Only)
Sea Level 59 °F (15 °C) Standard Day:	
Fuel Consumption	202 lb/hr (20 Shaft horsepower 84lb/ min Bleed Flow)
Output Shaft Horsepower	60 HP
Rotor Speed	100% (64,154 PRM)
Rotor Overspeed	105% (67,362 RPM)
Rotor Underspeed	95% (54,531 RPM)
Exhaust Gas Temperature (EDT):	
Start Acceleration Steady State	1,890 °F (1,032 °C) 1,324 °F (717 °C)
Oil Pressure	35 PSIG (minimum) 60 PSIG (maximum)
Oil Temperature	275 °C (135 °C)(maximum)
Oil Type	MIL-L-7808 (-54 to +54 °C (-65 to +130 °F)) MIL-L-23699 (-40 to +54 °C (-40 to +130 °F))

Table 49-1. Leading Particulars

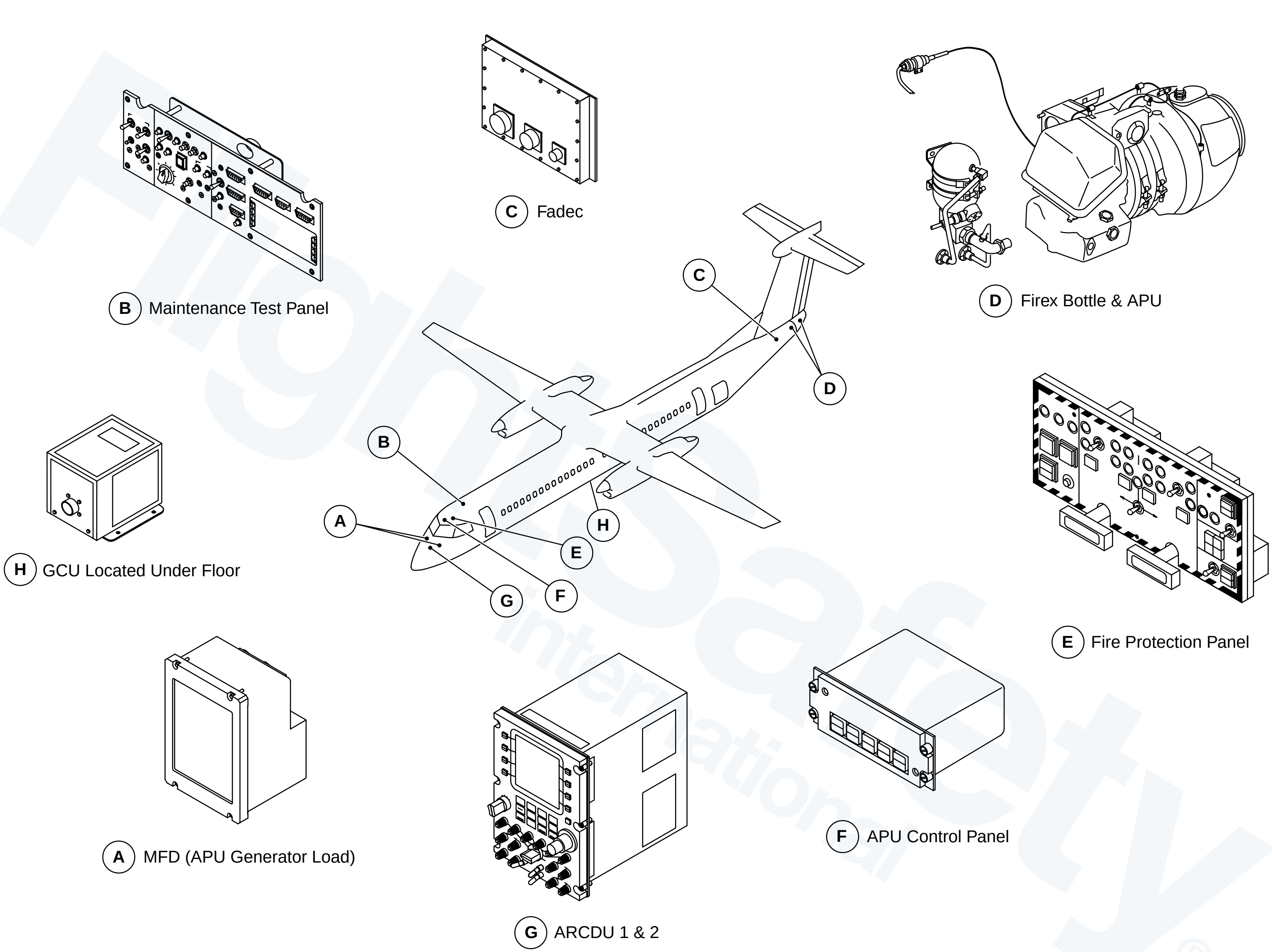


Figure 49-2. APU and Related Components

SYSTEM DESCRIPTION

Refer to Figure 49-2. APU and Related Components.

The FADEC controls, monitors and diagnoses all phases of the APU system operation and communicates fault information to the CDS for display on the ARCDU's when APU PWR selected is ON.

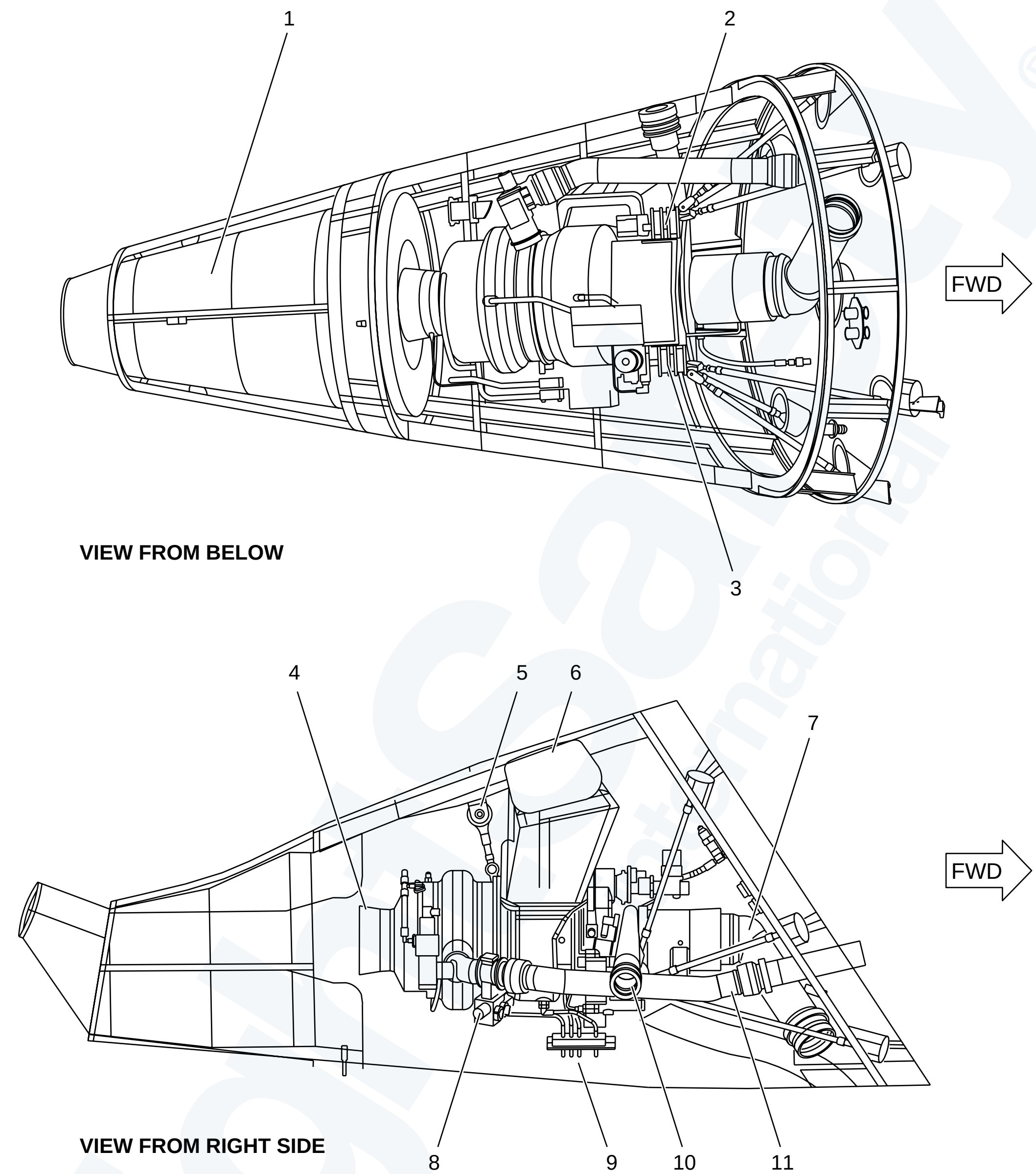
The FADEC controls the APU start, fuel flow, ignition, and shutdown. The FADEC also monitors engine speed, EGT, oil temperature, oil pressure, and the ambient pressure.

Abnormal operations are recorded in the FADEC's non volatile memory and a data memory module.

The air inlet is located on the right side above the APU access doors. The air induction system provides outside air to the APU compressor. Some of the compressed air is bled off the APU to be delivered to the aircraft Environmental Control System.

The gearbox assembly is a one piece aluminum casting with mounting pads for the engine and gearbox drive accessories.

NOTES



LEGEND

1. Exhaust Silencer
2. Right Side Mount
3. Left Side Mount
4. APU Exhaust
5. Rear Mount
6. Inlet Duct
7. Generator Inlet Duct
8. Bleed Valve
9. Drain Mast
10. Generator Outlet Duct
11. Bleed Air Duct

Figure 49-3. APU Powerplant View

49-10-00 POWERPLANT

GENERAL

Refer to:

- Figure 49-3. APU Powerplant View.
- Figure 49-4. APU Access and Lower Intakes.

The APU is located in a fireproof titanium compartment which forms the tail cone of the aircraft. Access to the APU is through two hinged doors that are part of the tail cone structure.

The APU is mounted to the empennage structure by a three point mounting system. The two forward mounts are attached to the gearbox and the rear mount is attached at the top of the APU. A fish pole hoist is used to install and remove the APU. The hoist is attached to a lug in the APU compartment and then connects to the APU front and rear lifting brackets.

The APU can be cradled onto a work stand for maintenance.

WARNING

MAKE SURE YOU HAVE SUFFICIENT PERSONS TO REMOVE OR INSTALL THE APU. THE APPROXIMATE WEIGHT OF THE APU IS 100 LBS (45 KG).

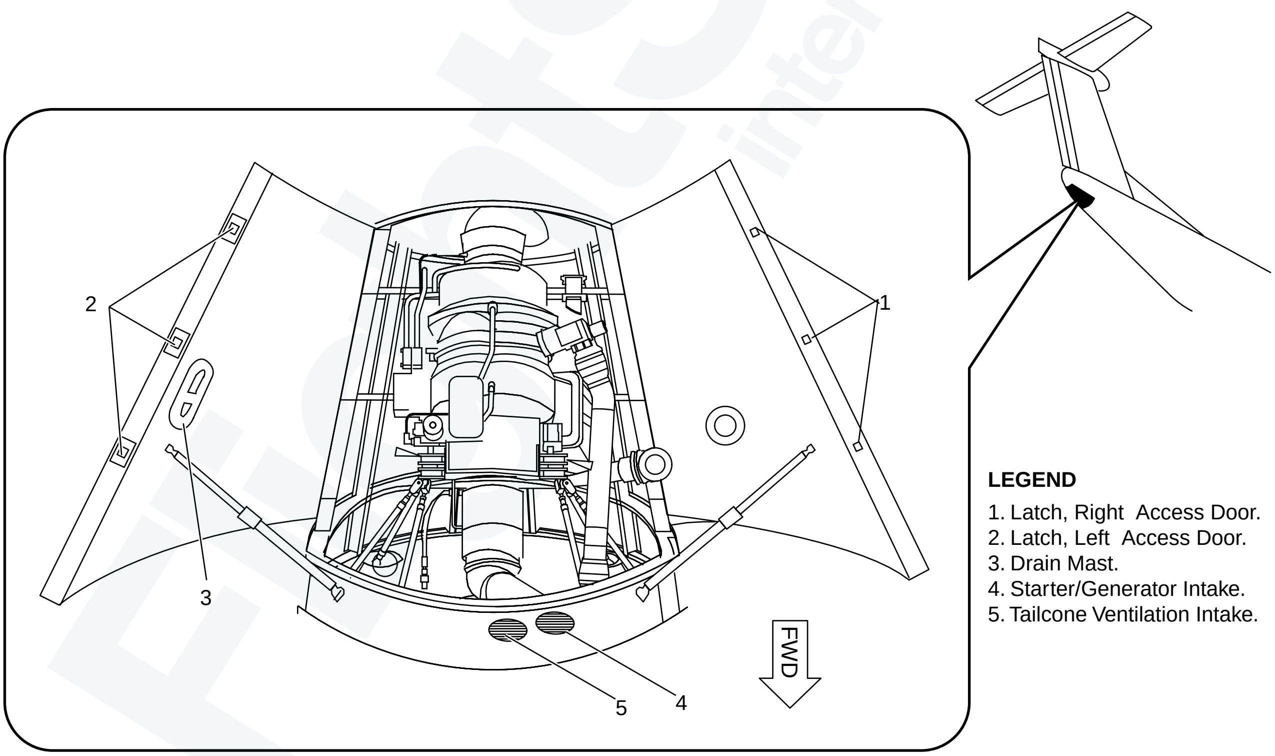


Figure 49-4. APU Access and Lower Intakes

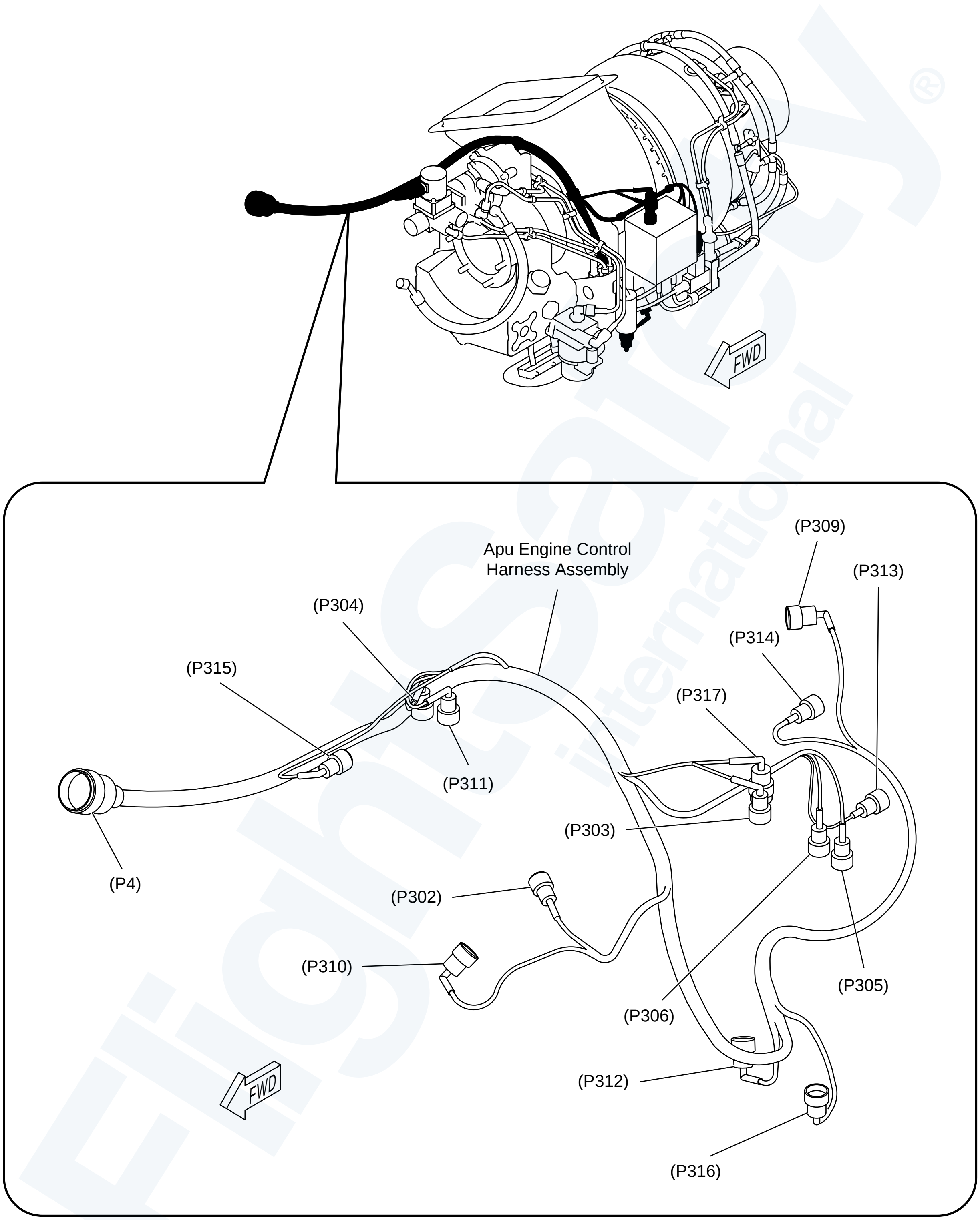


Figure 49-5. APU Electrical Harness

COMPONENT DESCRIPTION

APU Electrical Harness

Refer to:

- Figure 49-5. APU Electrical Harness.
- Figure 49-6. APU FADEC Connections.

The engine control harness assembly consists of 15 electrical connectors and associated connecting wires. The engine control harness connection is on the firewall bulkhead.

The electrical interconnect harness is installed from the firewall bulkhead to the APU FADEC in the rear equipment bay.

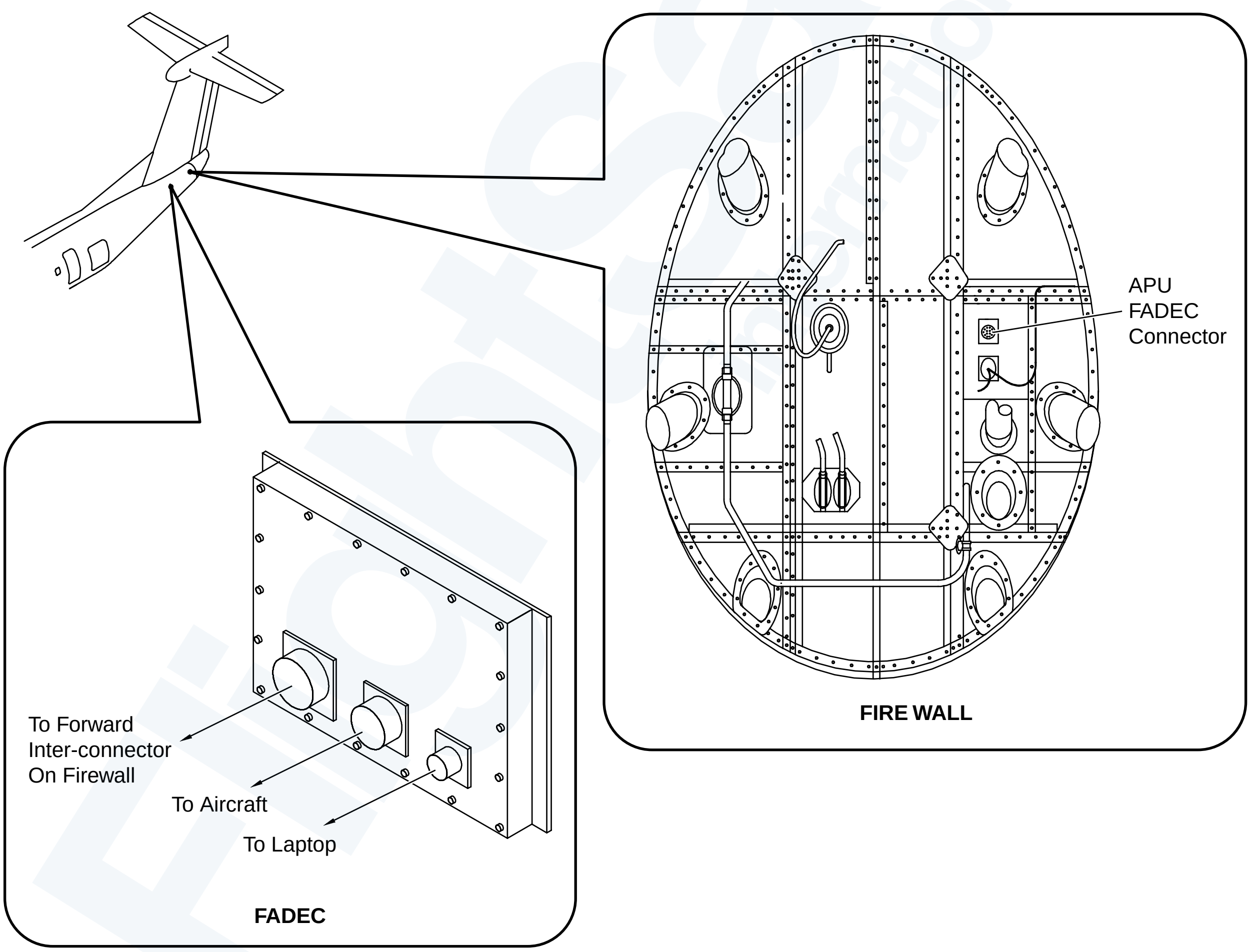


Figure 49-6. APU FADEC Connections

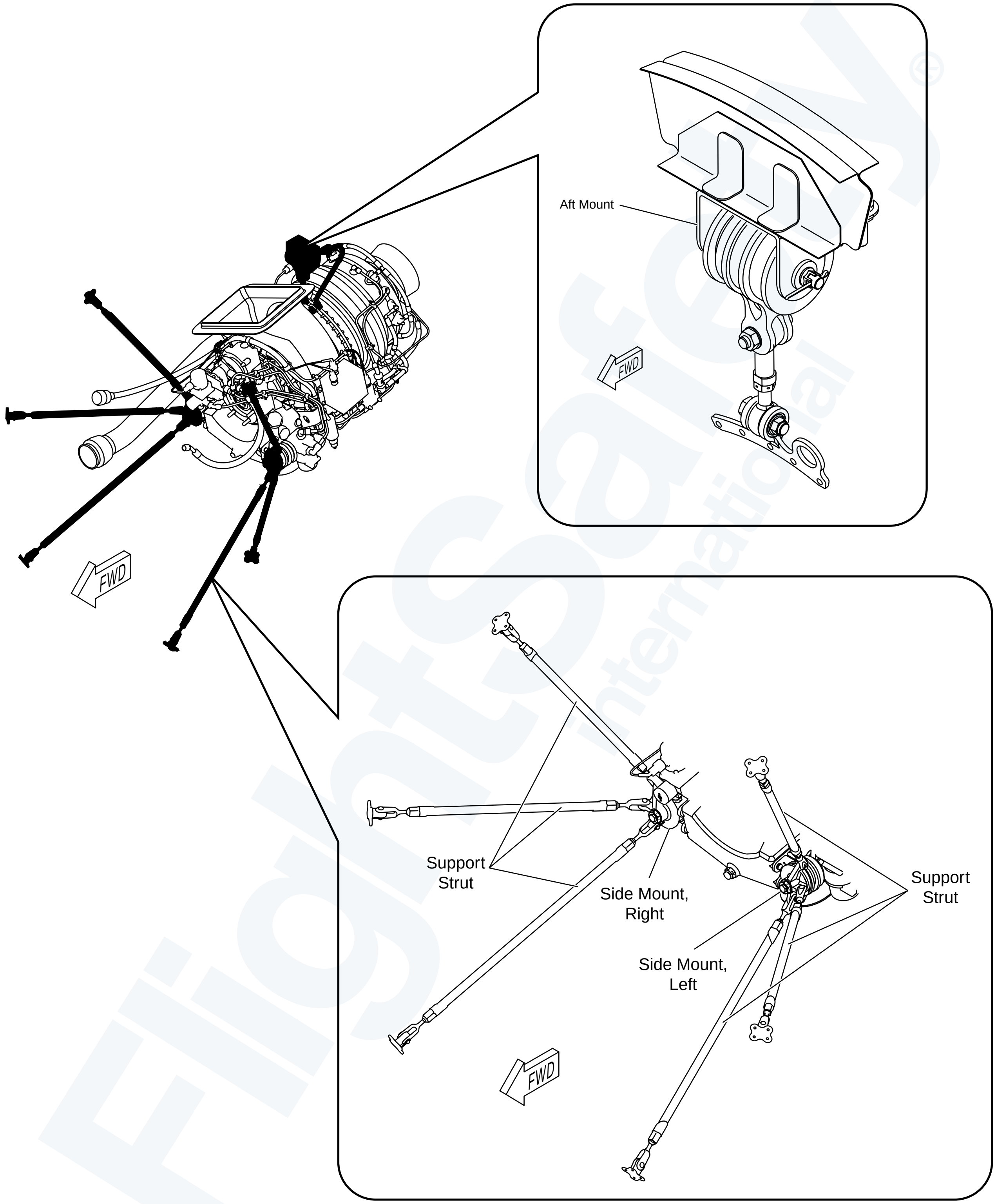


Figure 49-7. APU Front and Top Mounts

Mounts

Refer to Figure 49-7. APU Front and Top Mounts.

Mounts are used to support the APU and give vibration isolation to the aircraft structure.

The APU mounting system is designed to support the APU loads with any single component failure. In the event of fire, the elastomer maybe melted but the mount will snub and the APU location will be held safely.

The mounting system includes the components that follow:

- Side mount struts (6)
- Gearbox mounts (2)
- Aft mount
- Aft strut.

The aft APU mount is attached to the APU at the top of the turbine combustor and to the lower side of the center upper beam in the tailcone structure.

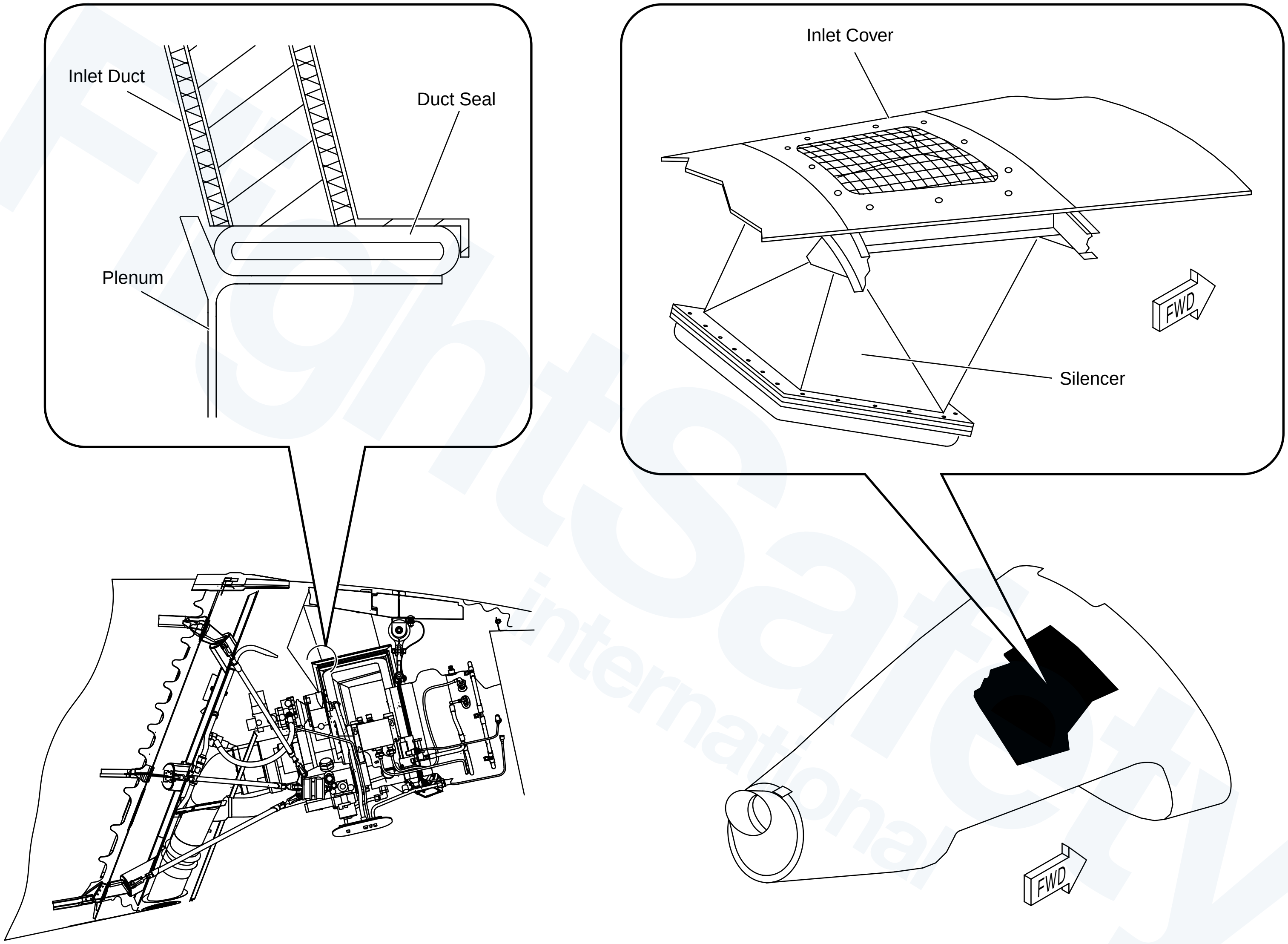


Figure 49-8. APU Right Side Air Intake

Refer to:

- Figure 49-8. APU Right Side Air Intake.
- Figure 49-9. Inlet Duct.

Protective Screen Cover

(No Longer a Louvered cover) Inlet is protected by a mesh screen, which can be replaced by a louvered door for winter operations. Installed on the right side of the tailcone to prevent Foreign Object Damage (FOD).

Silencer

The duct is a fireproof, acoustically treated, bonded assembly. Installed in the top right side of the forward tailcone, between the protective screen and the plenum of the APU.

Fireproof Seal

There is a fireproof silicone rubber seal where the duct attaches to the APU. This seal prevents tailcone air from entering the APU inlet system. It also allows for minor misalignment and motion between the APU and the structure. Installed between the Silencer and the APU inlet.

CAUTION

BE CAREFUL NOT TO CAUSE DAMAGE TO THE AIR FLOW DUCT DURING THE REMOVAL OR THE INSTALLATION OF THE APU.

Figure 49-9. Inlet Duct

APU Drains

Refer to Figure 49-10. APU Drains.

Drain connections are supplied for the components that follow:

- Fuel pump drive
- Gearbox breather
- Inlet plenum
- Combustor housing.

The stainless steel tubes connect to the drain mast. The drain mast is made of titanium; is airfoil shaped and protrudes through the left door of the tailcone.

All intercostals and frames in the tailcone and doors, have drain cutouts or holes at the lowest point to prevent the accumulation of liquids in the tail cone.

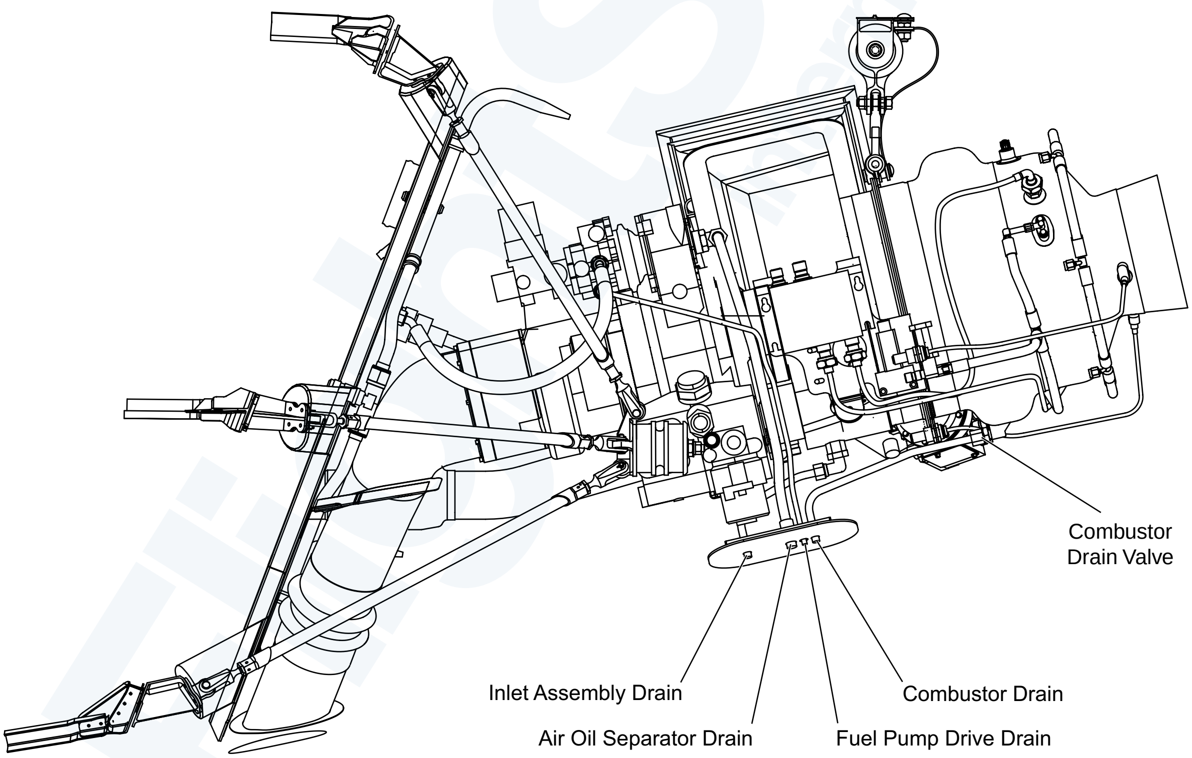


Figure 49-10. APU Drains

Generator Inlet and Outlet Ducts

Refer to Figure 49-11. Generator Inlet Duct.

Ambient air is drawn through a screened access hole on the lower fuselage forward of the Clamshell doors directly to the generator. After the air passes through the generator the hot air is exhausted through the outlet duct back to ambient.

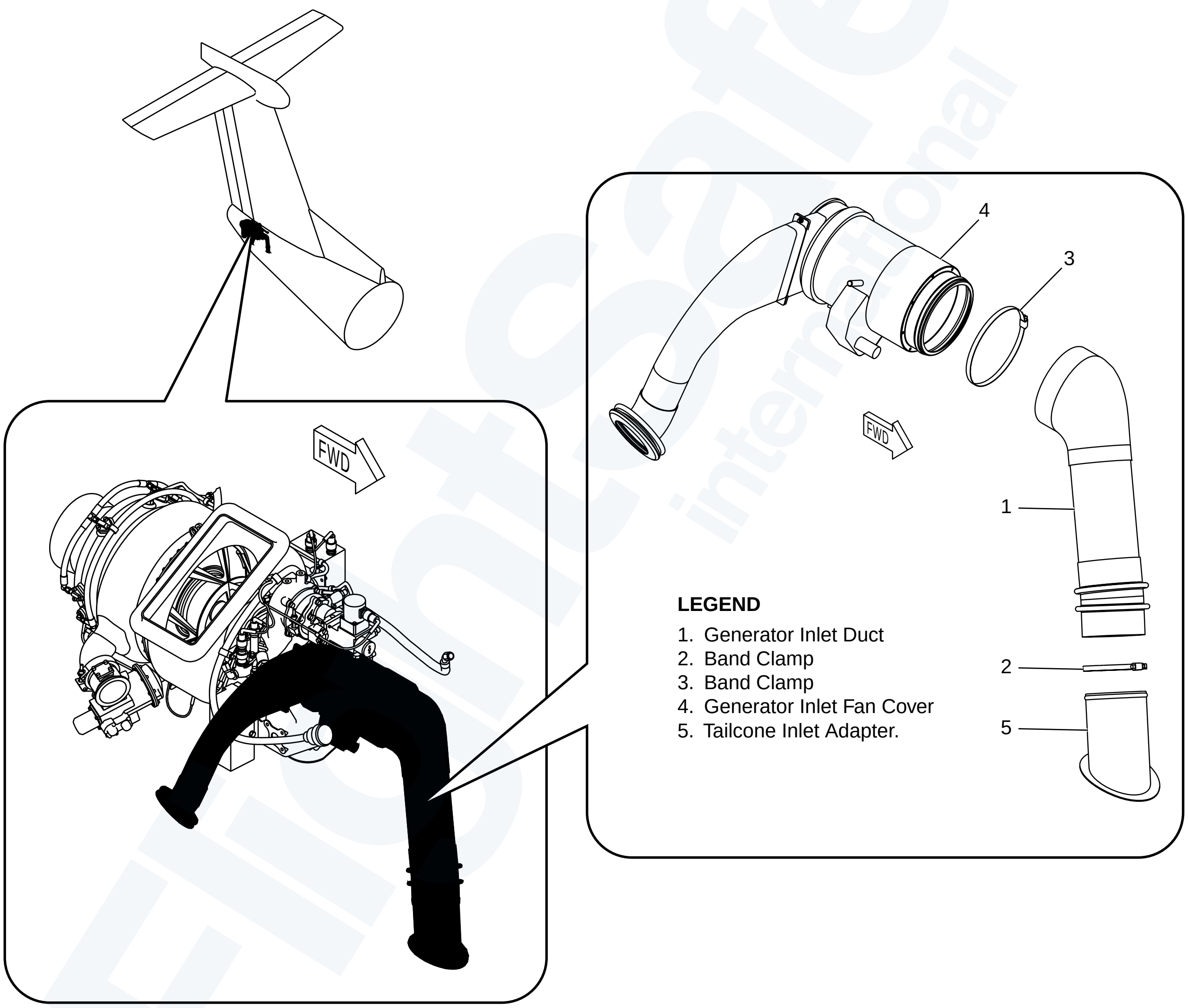


Figure 49-11. Generator Inlet Duct

TROUBLESHOOTING

Operational Test of the Auxiliary Power Unit

The following is an abbreviated descriptionof the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

If the APU FADEC or DMM has been changed, select the power switchlight on for a minimum 30 minutes to allow the memory buffers to synchronize.

The operational test requires two persons, one in the flight compartment and a second at the APU to monitor conditions.

With the APU running, visually check for fluid and air leaks, unusual noise smoking or exhaust flame. Check the APU for speed oscillation, no oscillation is allowed under any operating condition.

Functional Test of the Auxiliary Power Unit

The following is an abbreviated descriptionof the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Operate the APU for 5 minutes with no load. Visually inspect the APU.

Operate the APU BL AIR for a minimum 3 minutes, make sure the OPEN light comes on.

Select the GEN switchlight to ON and make sure that the ON light comes on.

NOTES

49-20-00 ENGINE

GENERAL

Refer to Figure 49-12. APU Engine Modules.

The Engine Assembly is a gas turbine engine with a single-stage centrifugal compressor, an annular combustor and a single-stage radial inflow turbine. A FADEC controls operation of the Engine Assembly.

The engine assembly has the following three modules:

- Turbine Module
- Combustor Module
- Gearbox Module.

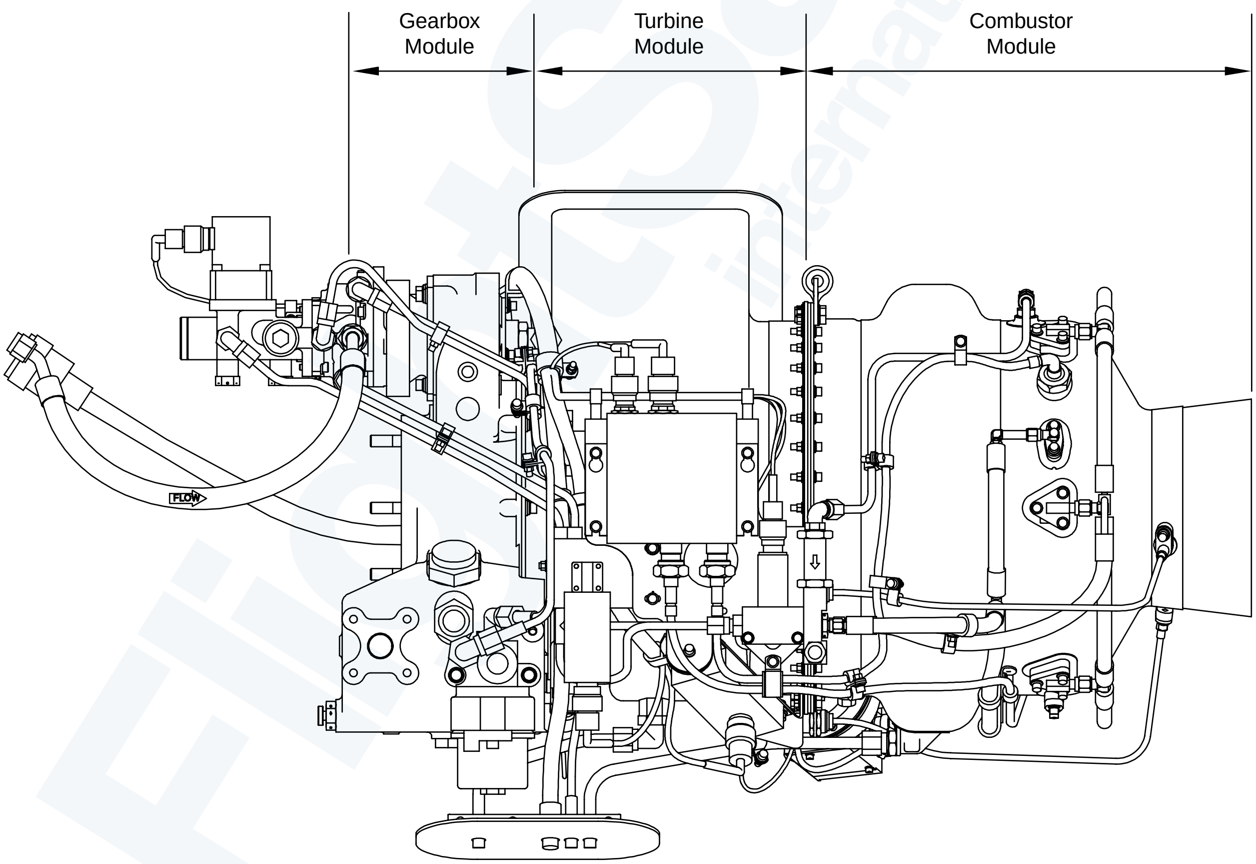


Figure 49-12. APU Engine Modules

COMPONENT DESCRIPTION

Turbine Module

Refer to Figure 49-13. Combustor and Turbine Assemblies.

The turbine module has the following components:

- Air Inlet Housing
- Rotor Assembly
- Turbine Nozzle Assembly.

Air Inlet Housing

The Air Inlet Housing assembly supplies the air to the APU compressor. The air inlet cast inconel housing provides a rigid support between the gearbox and combustor assemblies and provides containment for compressor rotor burst.

Balanced Rotor Assembly

The single stage rotor is a balanced assembly. It consists of a radial compressor wheel and turbine wheel positioned on a two bearing shaft system. A stationary segmented seal plate between the compressor and turbine wheel prevents compressor discharge air from bypassing the combustor.

Turbine Nozzle/Containment Assembly

The assembly has a containment ring and a turbine nozzle. The turbine nozzle is attached to the containment ring by seven radial pins. Four bolts that pass through the diffuser and the air intake housing secure the complete assembly on the forward side of the air intake housing. Inside the turbine nozzle are 21 vanes, which direct the high velocity gas stream to the turbine wheel. Hot gas flows discharge axially through the turbine exhaust. Rotational energy from the hot gas flow to the turbine wheel, drives the compressor and the generator.

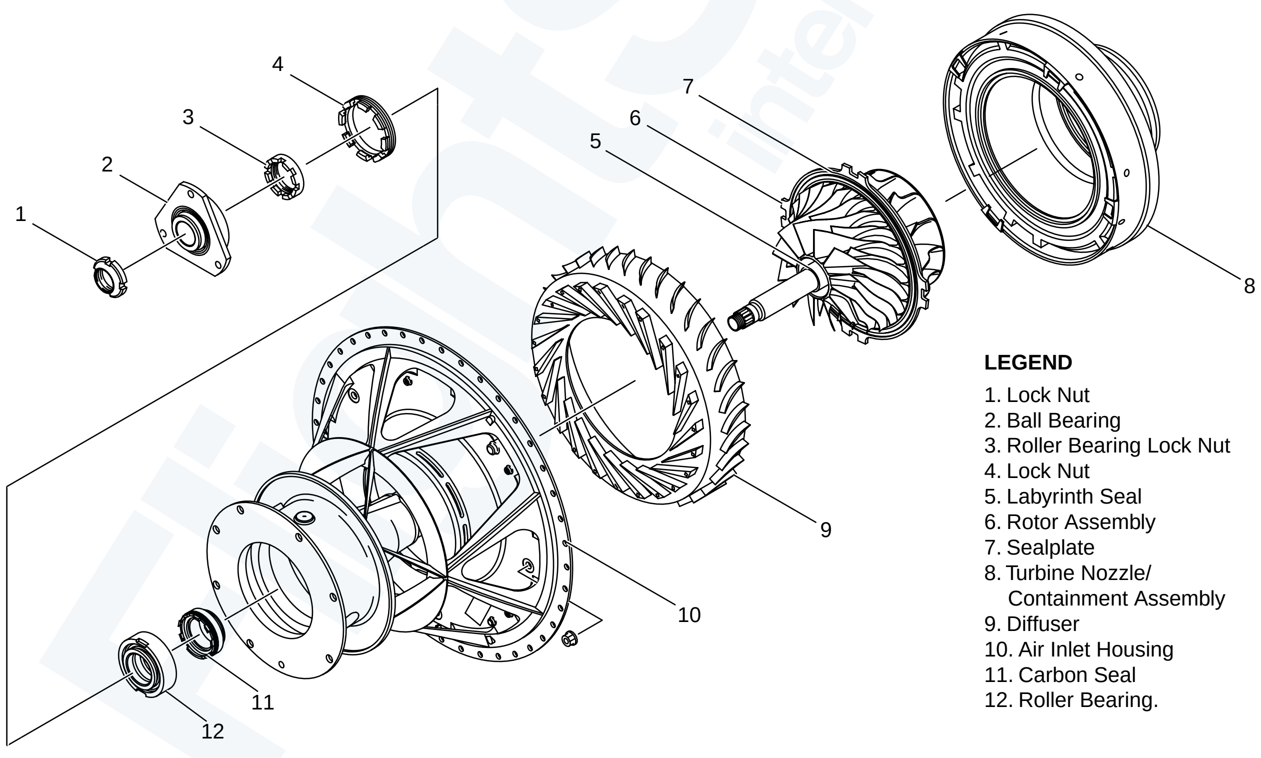


Figure 49-13. Combustor and Turbine Assemblies

COMBUSTOR MODULE

Refer to Figure 49-14. Combustor Module.

The combustor module includes the following components:

- Combustor Housing
- Combustor Liner.

The combustor housing is attached to the air inlet housing with bolts. The Combustor Module consists of the combustor housing, combustor liner, start and main fuel nozzles, and fuel manifold. A 2.5 inch bleed port is provided on the right side of the combustor housing for ECS bleed air. Compressed air from the impeller is discharged into the combustor housing which forms a plenum surrounding the combustion chamber liner. Air enters the combustion chamber liner through dilution holes.

Combustion takes place in the annular combustor chamber liner. Dilution air is added to produce a uniform flame and the hot gas stream. Two start nozzles, six main fuel injectors, two ignitors and one radial pin help to locate the combustion chamber in the housing.

The combustor drain valve is threaded into the combustor drain boss. It makes sure any accumulation of fuel in the combustor is drained overboard through the combustor drain system upon shutdown (<3 psi P3 pressure).

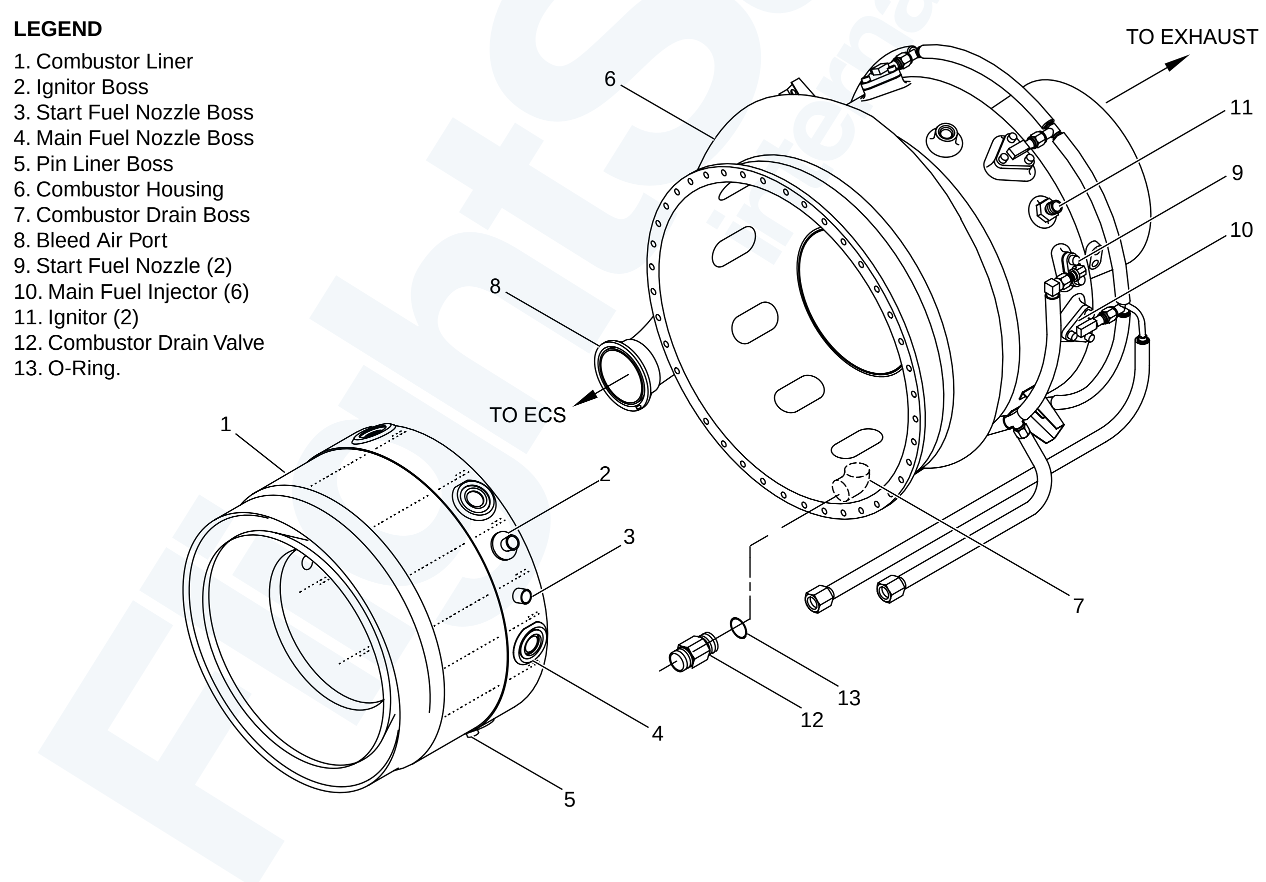


Figure 49-14. Combustor Module

Gearbox Module

Refer to Figure 49-15. Gearbox Module Gear Rotations.

The gearbox housing is a machined aluminum casting with a fireproof oil sump. All gear shafts are supported in the housing by roller and ball bearings. The aft section of the gearbox provides a flange for the attachment to the engine turbine module. The gearbox has a main output pad for the starter/generator and a second pad for the fuel control/pump assembly. Bosses on the housing allow for the mounting of high oil temperature and low oil pressure switches.

The gearbox module provides the reduction of the turbine rotor speed to speeds required for starter/generator, fuel pump, and oil pump drives. The input quill drives a set of fixed star gears, which transfer power to a ring gear. The oil pump gear is driven from the output shaft. The fuel pump gear is driven through a single idler gear. The fuel pump gear includes an air/oil separator for gearbox venting.

Gearbox Vent Seal

The gearbox vent seal is installed on the top rear of the gearbox between the fuel control unit and gearbox.

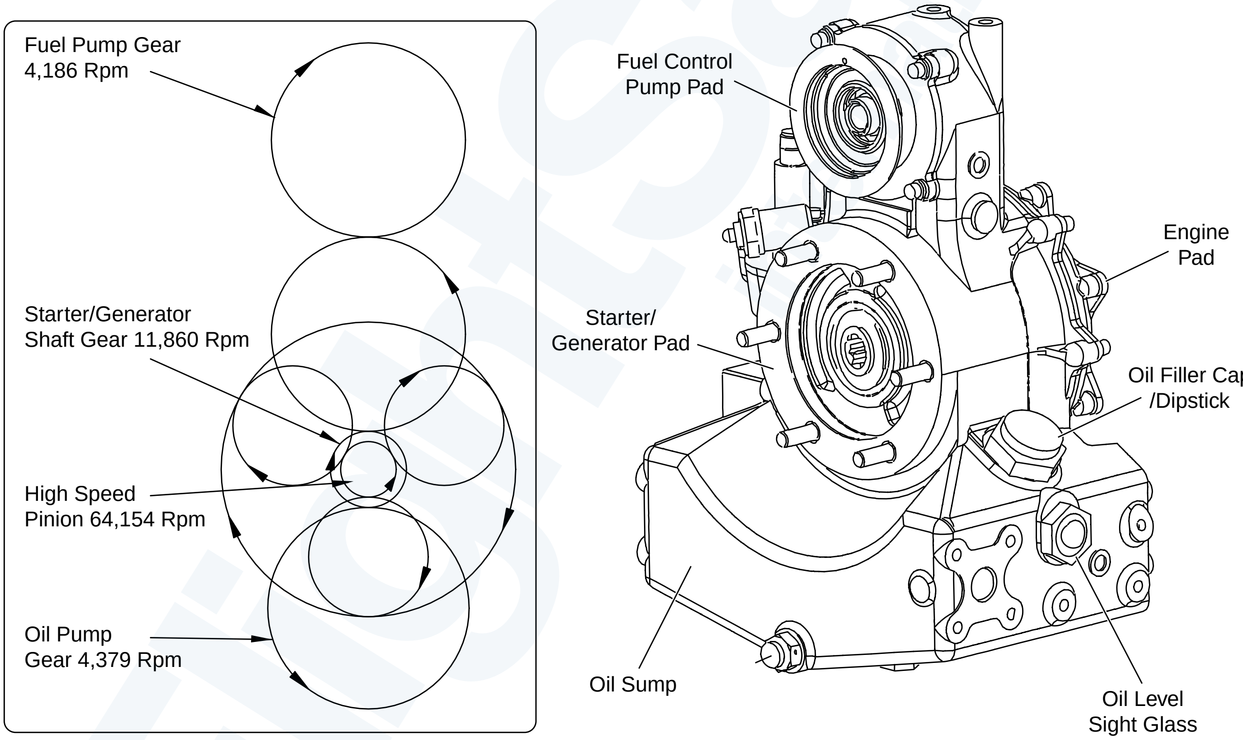


Figure 49-15. Gearbox Module Gear Rotations

OPERATION

The gas turbine APU provides for a combination of shaft horsepower and compressor bleed air.

Ambient air flows through a mesh screen then into the inlet of APU compressor. The Rotor and Diffuser compress the ambient air and direct it into the combustion chamber. The chamber walls are cooled by air introduced from cooling strips that distributes the air over the inner surface of the chamber.

The six fuel injectors provide for complete combustion and even temperature distribution of the hot gas entering the turbine nozzle.

The one-piece turbine extracts the kinetic energy from the expanding gases and converts it into mechanical energy to rotate the turbine shaft. The turbine shaft uses the generated torque to rotate the gearbox. The hot spent gases flow axially to the APU exhaust duct.

The bleed port on the combustor housing sends bleed air to the ECS ducting when the APU bleed air valve is open.

NOTES

MAINTENANCE PRACTICES

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Removal of the APU Engine Assembly

Refer to:

- Figure 49-16. APU Engine Assembly - Removal/Installation (1 of 2).
- Figure 49-17. APU Engine Assembly - Removal/Installation (2 of 2).

Remove the following components from the engine:

- Generator inlet duct
- Generator exhaust duct
- APU DC generator
- Bleed duct
- Gearbox cooling duct
- Grounding strap.

Attach the APU lifting lug onto the APU engine assembly. Install the APU hoist. Attach the cable spring snaps to the forward and aft APU lifting lug.

Apply a light tension onto the APU hoist cables. Disconnect the three mounts. Lower the APU engine assembly on the APU build-up stand.

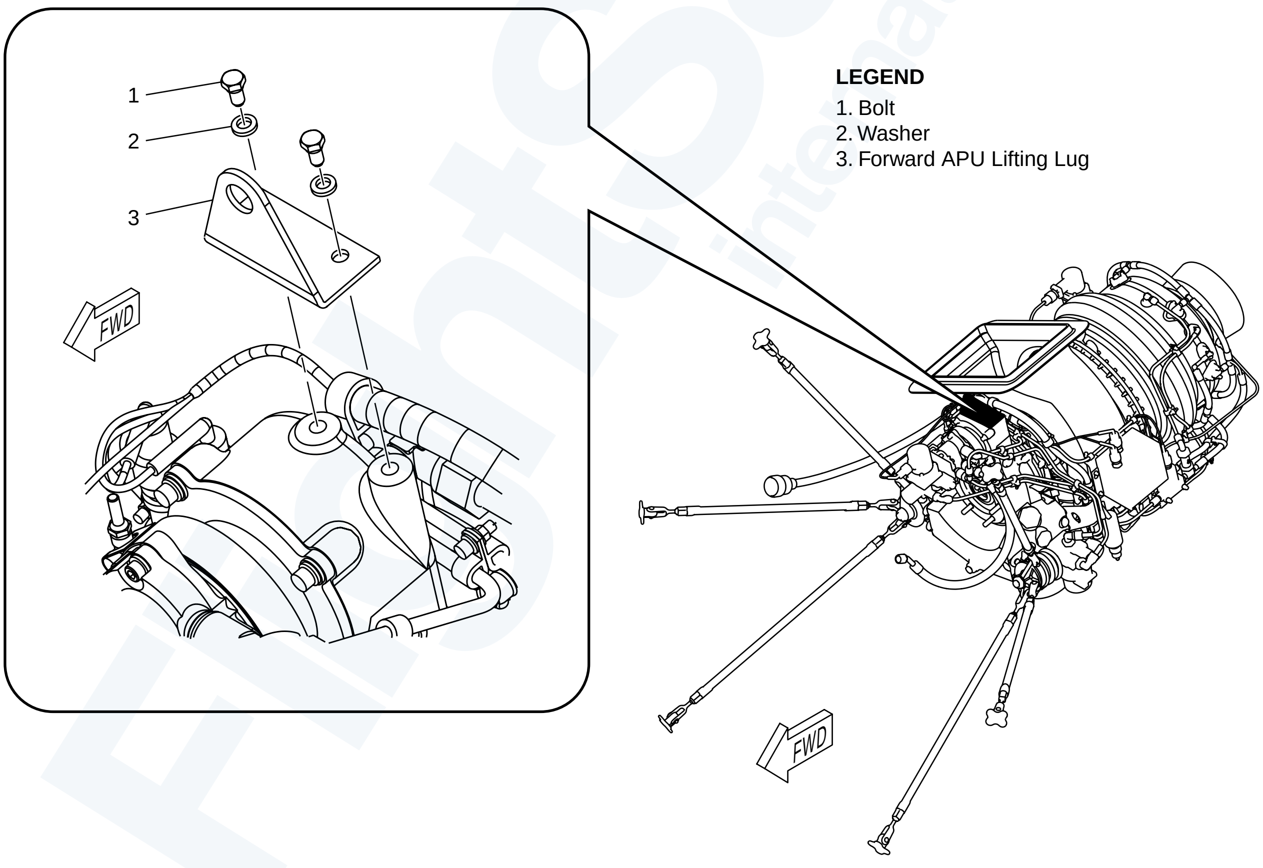


Figure 49-16. APU Engine Assembly - Removal/Installation (1 of 2)

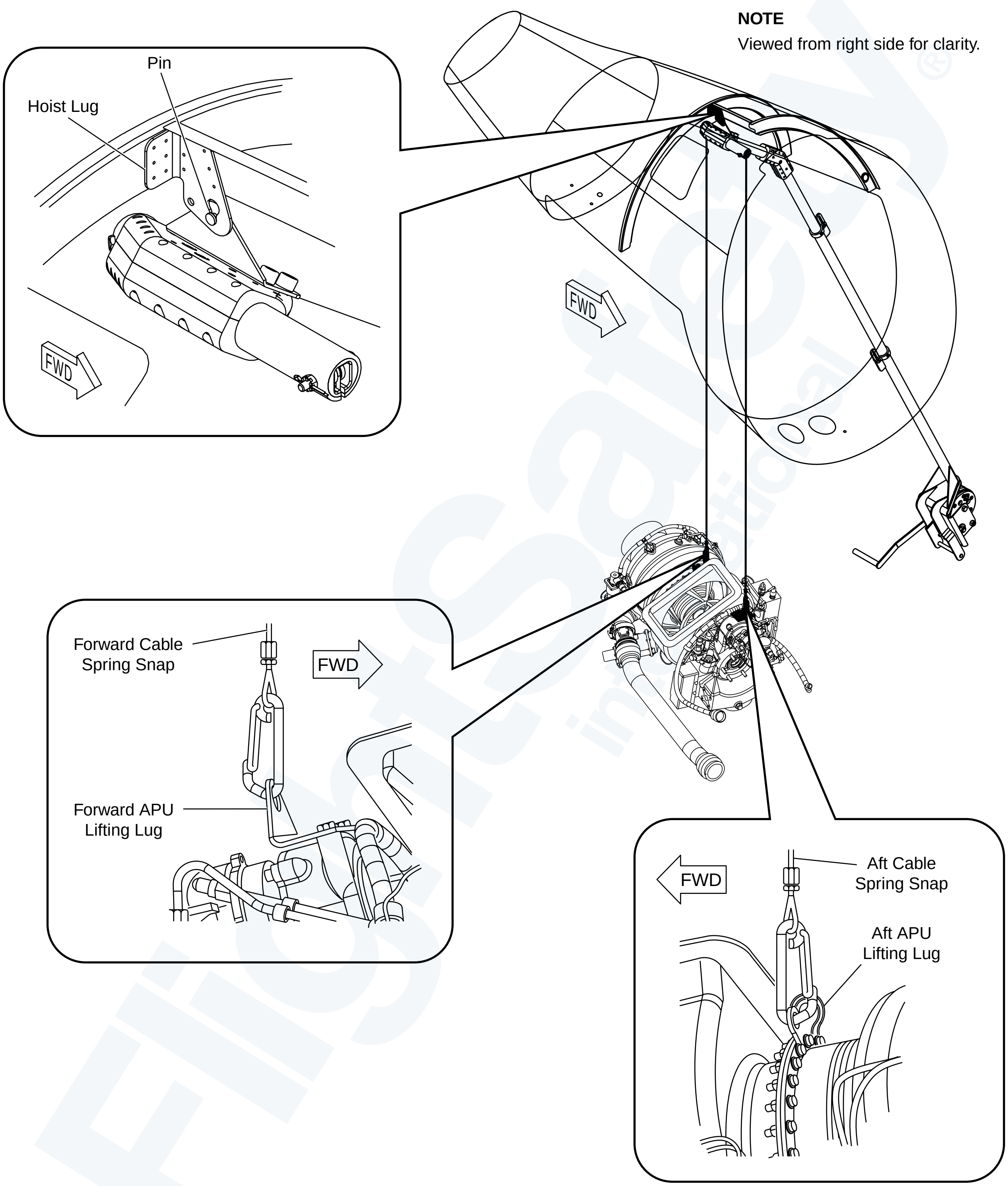


Figure 49-17. APU Engine Assembly - Removal/Installation (2 of 2)

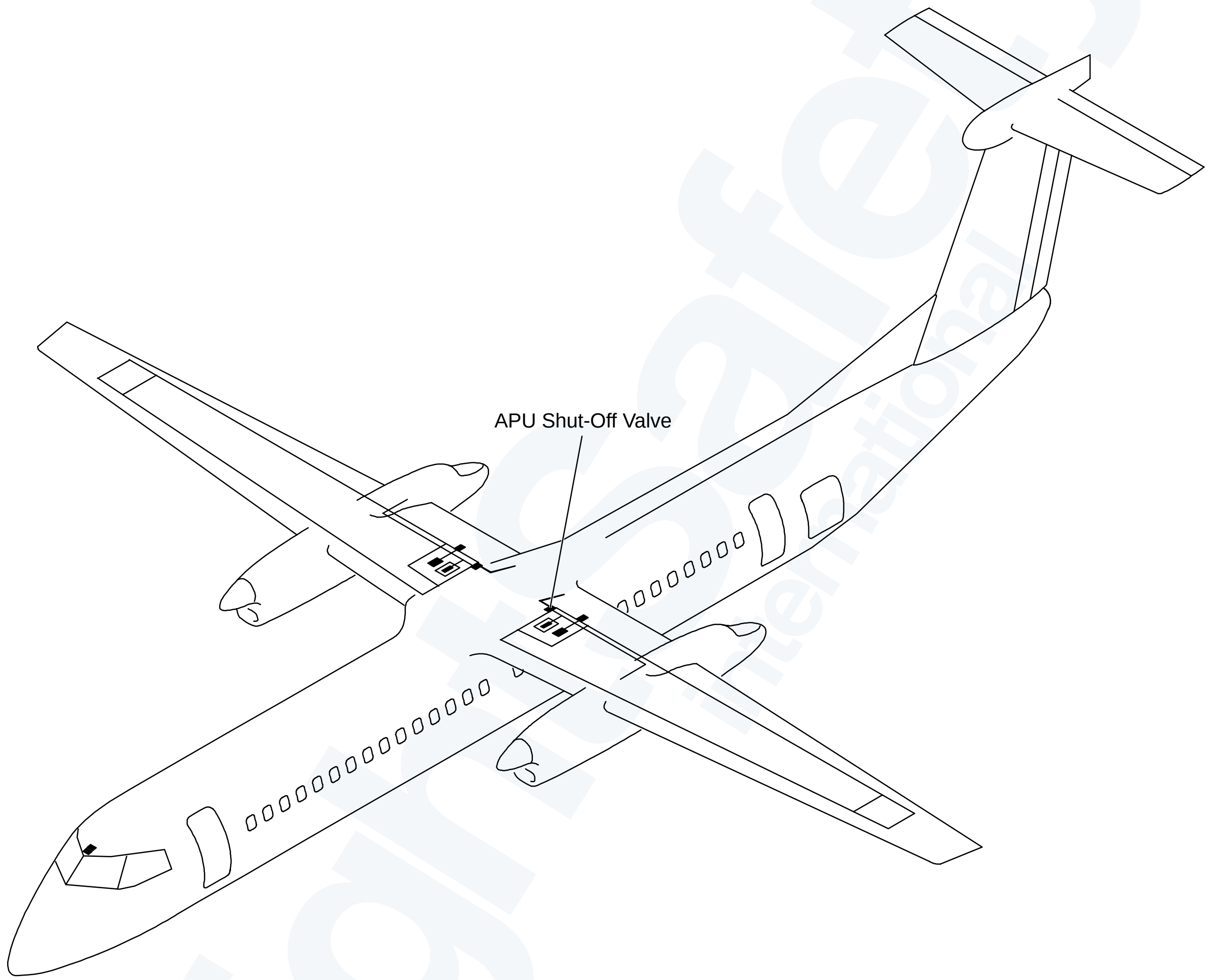


Figure 49-18. APU Fuel Feed Locator

49-30-00 ENGINE FUEL AND CONTROL

INTRODUCTION

The Engine Fuel and Control System delivers the fuel required to start and maintain a constant 100% RPM under varying load conditions and field elevations.

GENERAL

The APU fuel feed system supplies a flow of fuel from the (No.1) fuel tank to the optional APU.

Fuel to the APU is by gravity feed from the left wing fuel tank.

The APU fuel valve is opened and closed by the FADEC during normal start and stop operations.

The FADEC supplies the signals that operate the fuel control of the APU.

The system has the components that follow:

- FADEC
- Fuel control assembly
- Start fuel valve
- Main fuel valve
- Flow divider
- Purge check valve
- Start fuel nozzle
- Main fuel nozzle
- Start fuel manifold
- Main fuel manifold
- Fuel filter assembly
- Fuel filter element.

SYSTEM DESCRIPTION

Refer to Figure 49-18. APU Fuel Feed Locator.

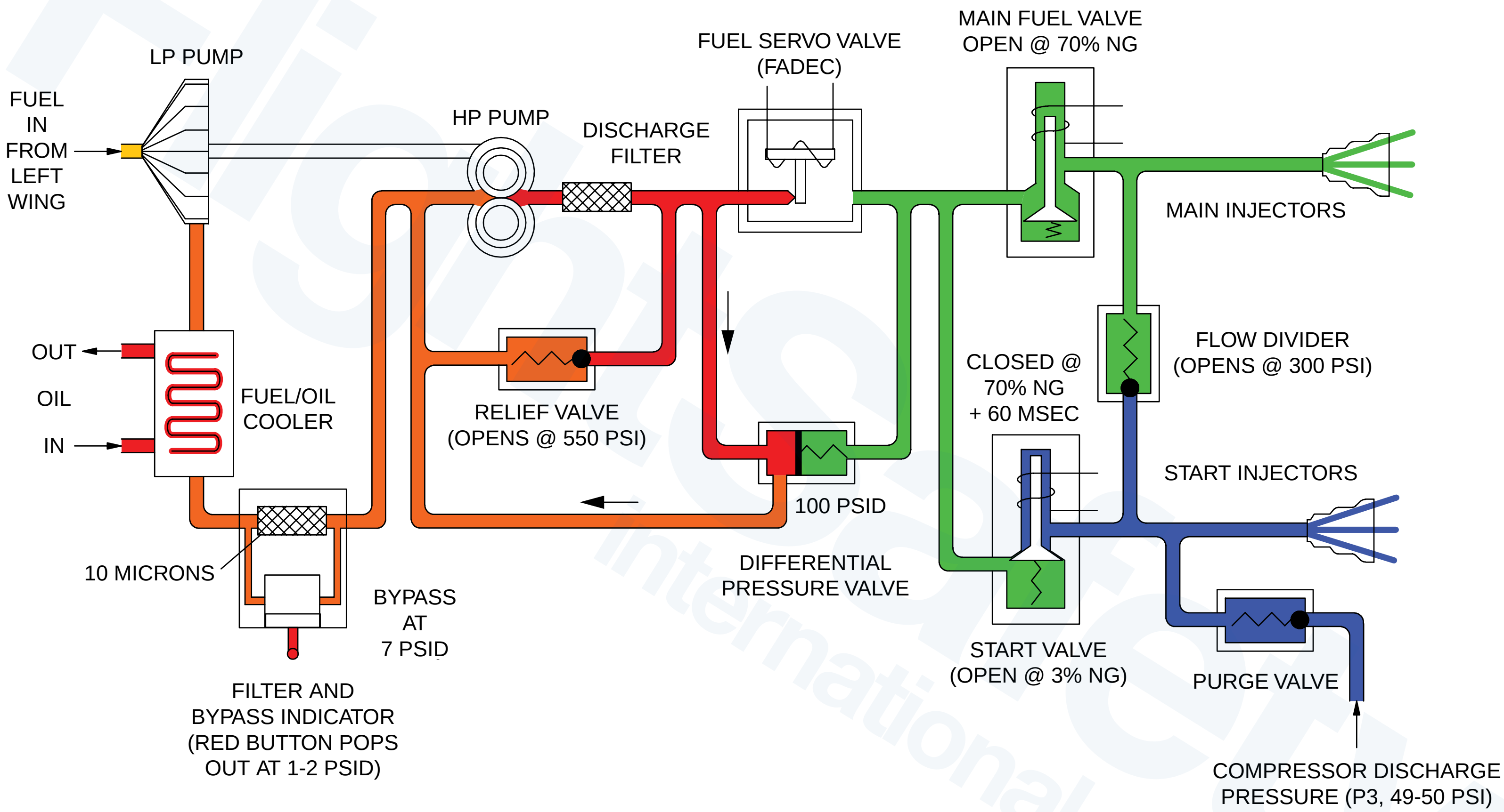


Figure 49-19. Fuel System Schematic

Refer to Figure 49-19. Fuel System Schematic.

The low pressure fuel pump receives fuel from the aircraft fuel system to supply the fuel/oil cooler, fuel filter, high pressure fuel pump, the fuel servo valve and the fuel solenoid manifold assembly. Excess fuel is circulated through the relief valve across the high pressure fuel pump. A differential pressure valve is in parallel with the fuel servo valve to maintain a constant differential pressure across the fuel servo valve. This results in a fuel flow linear relationship to the servo current command from the FADEC.

At start initiation, the FADEC records the Exhaust Gas Temperature (EGT), for reference during the start sequence. The FADEC uses speed sensor data for successive stages of the start sequence and to monitor APU operation for overspeed/underspeed conditions.

NOTES

COMPONENT DESCRIPTION

Fuel Control Assembly

Refer to Figure 49-20. Fuel Control Assembly.

The servo valve is energized open at start up. The fuel servo receives milliamp signals from the FADEC to maintain 100% Ng speed during all APU load conditions. In the shutdown sequence the FADEC maintains current to the fuel servo valve for 0.5 seconds and then de-energizes the valve.

The differential pressure valve is in parallel with the fuel servo valve to maintain a constant 100 psi differential pressure across the fuel servo valve. This results in a fuel flow linear relationship to the servo current command from the FADEC.

The fuel control assembly is mounted on the Gearbox module with a V-band clamp.

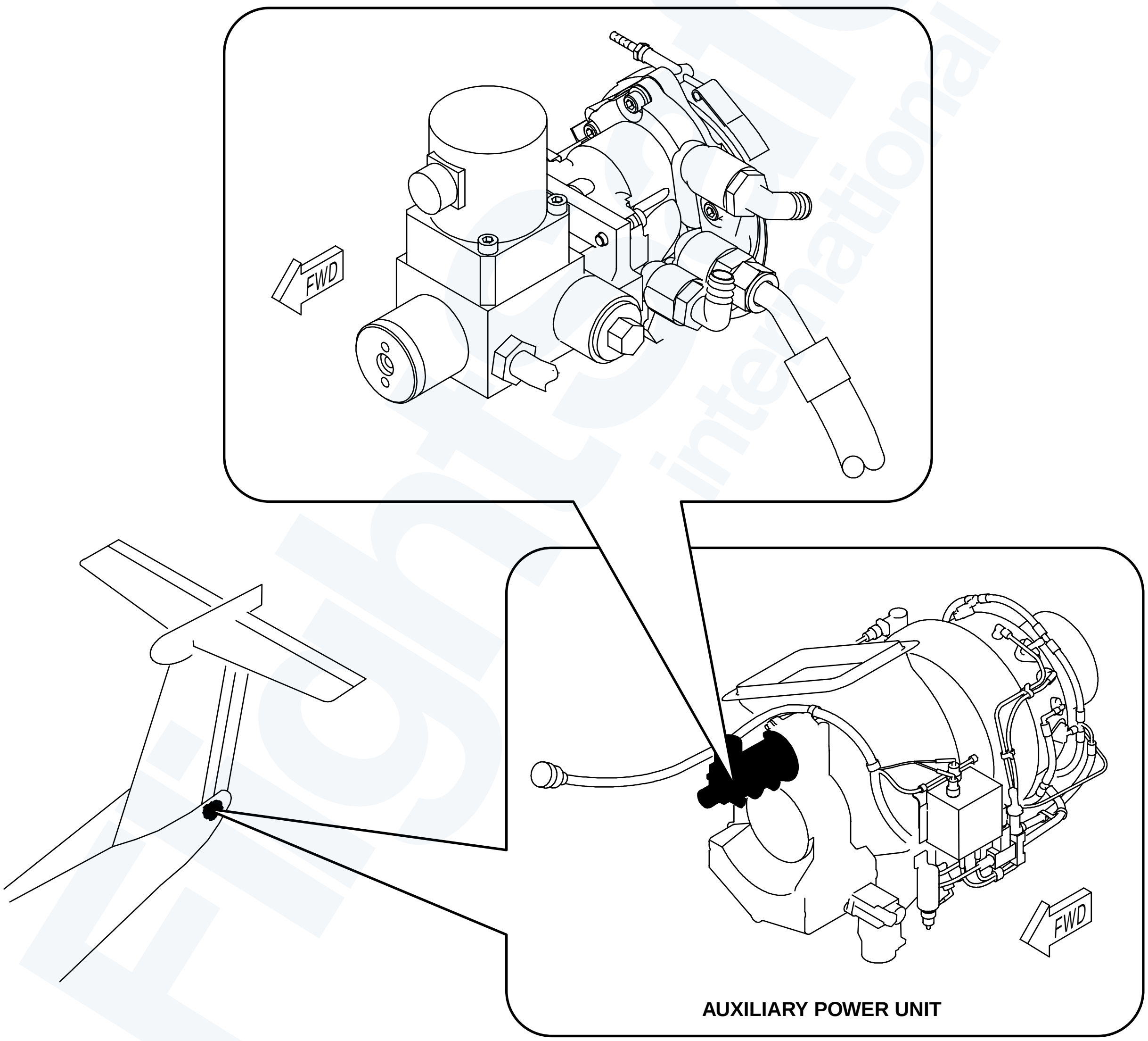


Figure 49-20. Fuel Control Assembly

Start Fuel Valve

Refer to Figure 49-21. APU Fuel Valves.

The start fuel valve is installed on the fuel solenoid manifold assembly. The valve is energized open at 3% Ng. The valve is closed at 70% Ng speed + 60 msec.

Main Fuel Valve

The main fuel valve is installed on the fuel solenoid manifold assembly. The valve is energized open at 70% Ng. The valve stays open during engine operation.

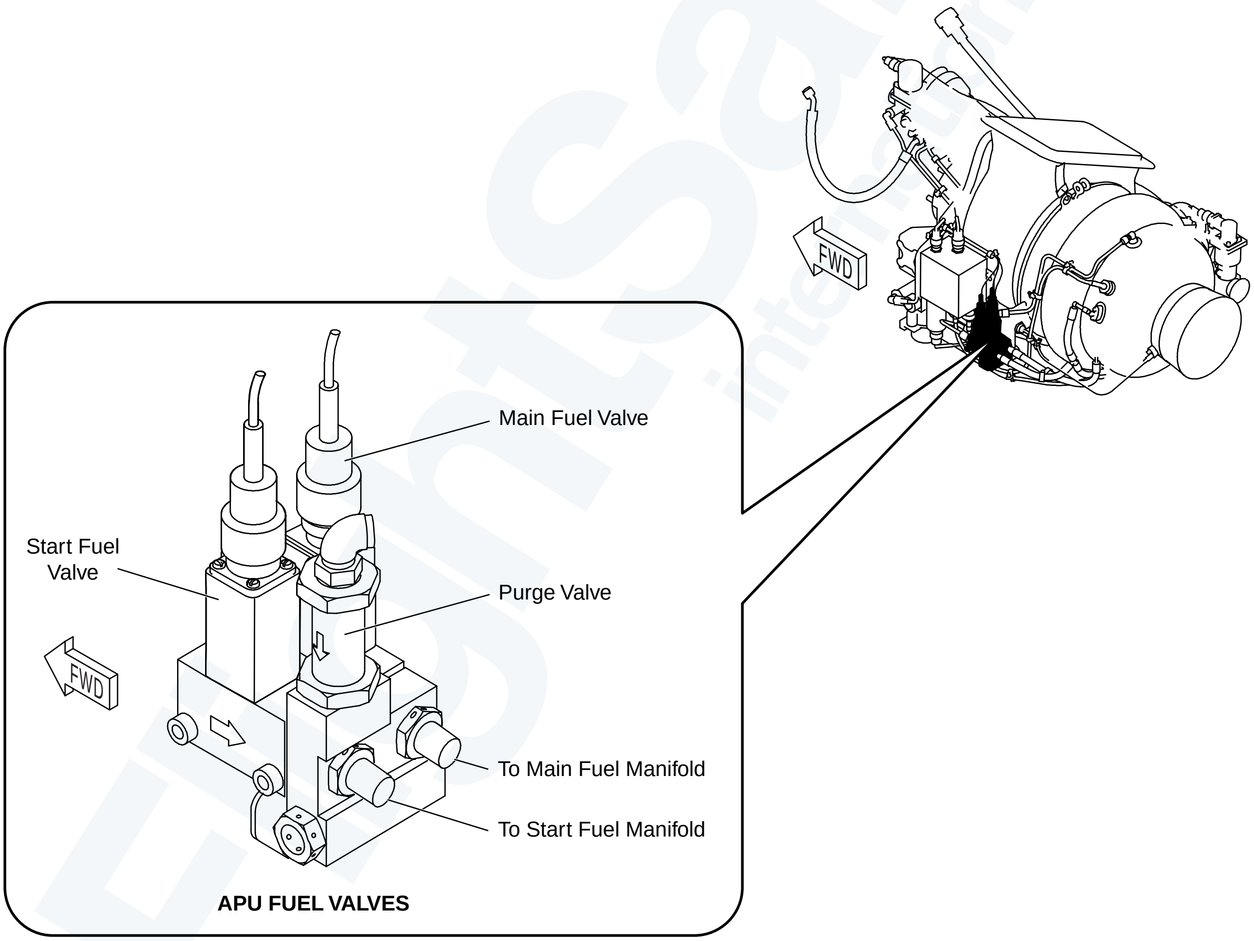


Figure 49-21. APU Fuel Valves

Combustor Drain Check Valve

Refer to Figure 49-22. Combustor Drain and Check Valve.

The valve is threaded into the combustor drain boss and is closed by air pressure in the combustor housing during engine operation. The Valve allows drainage of the combustor when the engine is not in operation.

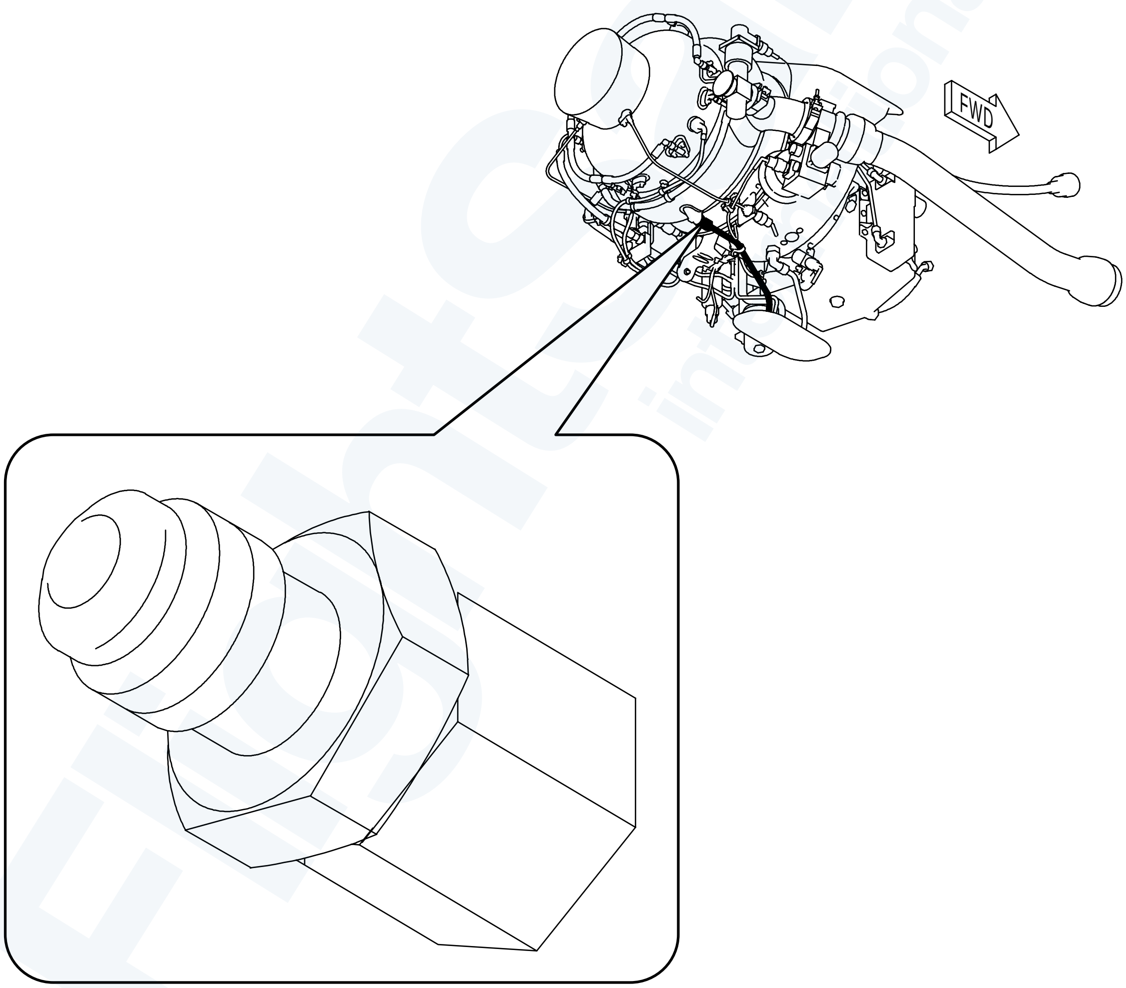


Figure 49-22. Combustor Drain and Check Valve

Flow Divider and Purge Check Valve

Refer to Figure 49-23. APU Flow Divider And Purge Check Valve.

The flow divider valve is located in the fuel solenoid manifold assembly. The valve opens when start fuel pressure reaches 300 psi (2068.4 kPa). This ensures proper fuel pressure is maintained to the start nozzles.

The purge check valve is spring loaded and is opened by the engine compressor air pressure when start fuel pressure is not present. The open purge check valve routes a continuous flow of air from the compressor to purge the residual fuel from the start fuel manifold and start nozzles. The air flow also cools and prevents the fuel from coking the start nozzles.

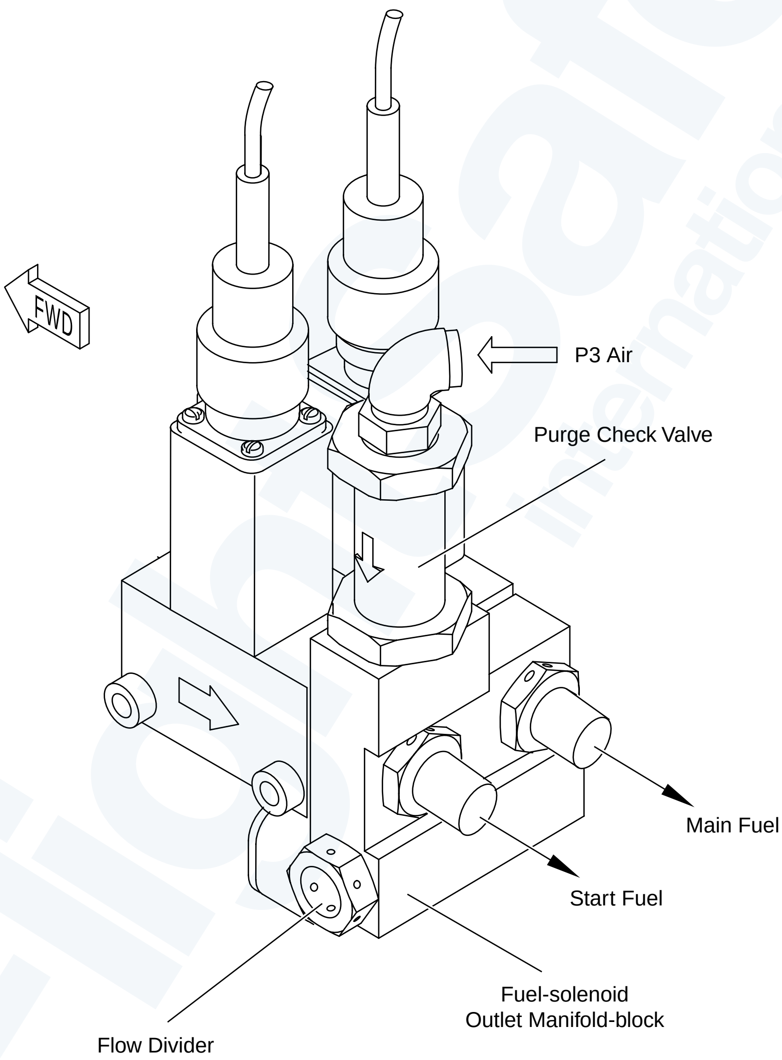


Figure 49-23. APU Flow Divider And Purge Check Valve

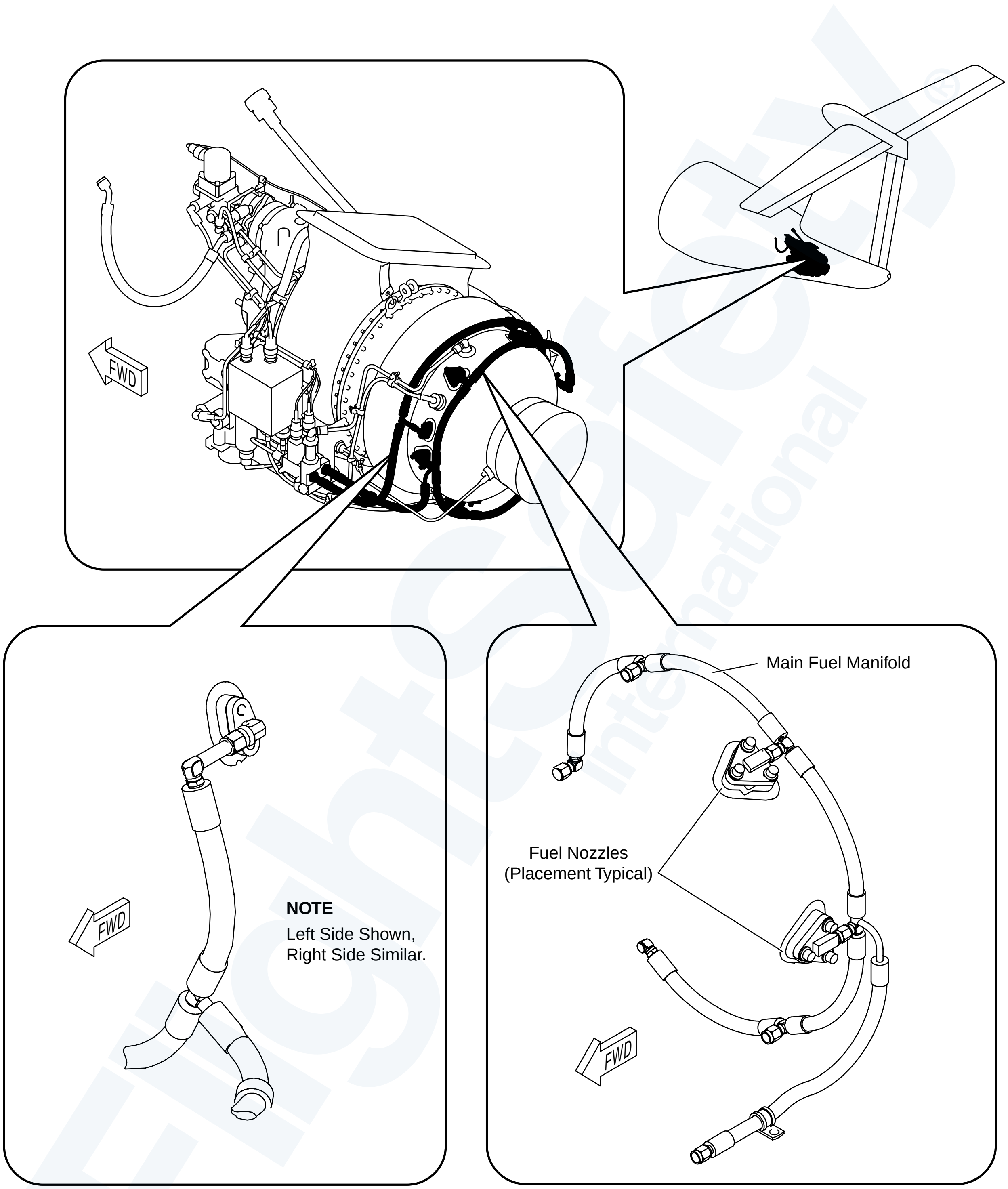


Figure 49-24. Start and Main Fuel Manifolds and Nozzles Locator

Refer to Figure 49-24. Start and Main Fuel Manifolds and Nozzles Locator.

Start Fuel Nozzle

The two start fuel nozzles are threaded into individual nozzle assemblies and installed in the combustor housing with bolts.

Start Fuel Manifold

The flexible start fuel manifold is located on the engine combustor housing and supplies fuel to the two start fuel nozzles.

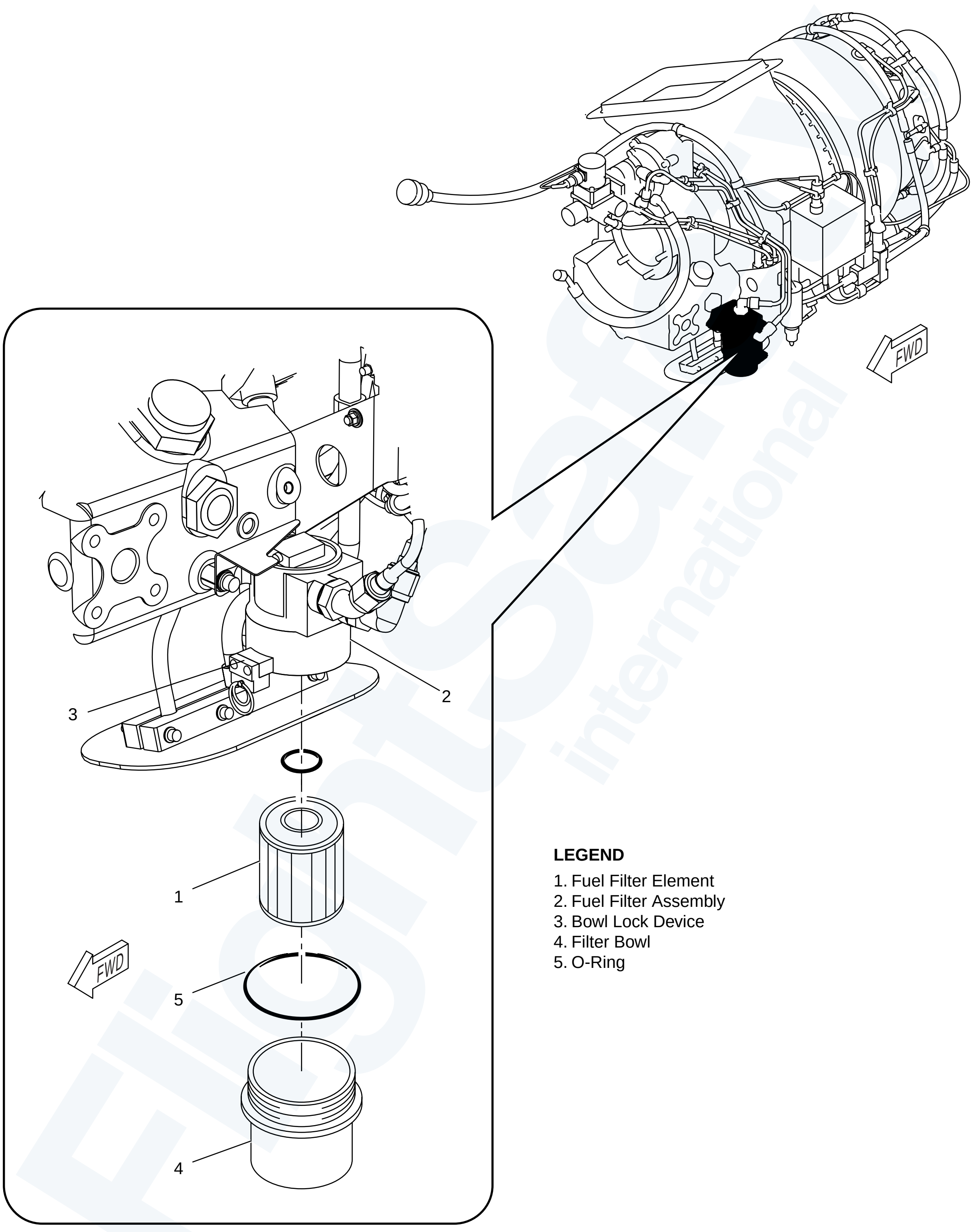
Main Fuel Nozzle

The six fuel nozzle assemblies are installed in the combustor housing with bolts.

Main Fuel Manifold

The main fuel manifold is flexible and is attached around the engine combustor housing. The manifold supplies fuel to the six fuel nozzle assemblies.

NOTES



LEGEND
1. Fuel Filter Element
2. Fuel Filter Assembly
3. Bowl Lock Device
4. Filter Bowl
5. O-Ring

Figure 49-25. Fuel Filter

Fuel Filter Assembly

Refer to Figure 49-25. Fuel Filter.

The filter differential pressure indicator and bypass valve are located on the side of the filter housing. The impending bypass indicator shows that the filter flow is restricted. The filter bypass valve opens at a pressure differential of 7 psid (48.3 kPad).

Filter Element

Located inside the filter assembly is a low-pressure 10 micron fuel filter that protects the fuel control unit from contamination.

NOTES

Data Memory Module

Refer to Figure 49-26. DMM and APS Sensor Locator.

The Data Memory Module (DMM) is a solid state unit located on the bottom side of the APU. The DMM stores APU operating hours, fault history (50 shutdowns and advisory faults), and the APU and FADEC serial numbers. The FADEC updates the DMM.

NOTE

The Data Memory Module serial number is the key to the FADEC check. The FADEC checks the serial number of the DMM. A different or new serial number tells the FADEC that the APU has changed. If there is a new DMM, the FADEC will load the DMM with data.

Ambient Pressure Sensor

The Ambient Pressure Sensor (APS) is mounted on the left side of the air inlet assembly and senses air inlet pressure to the engine. Pressure signals are sent to the FADEC for altitude correction of the fuel scheduling. After ready to load, which occurs at 95% plus 3 seconds, ambient pressure is used to set the minimum and maximum fuel set points. If the APS fails, the FADEC uses the preset standard day/ pressure altitude value.

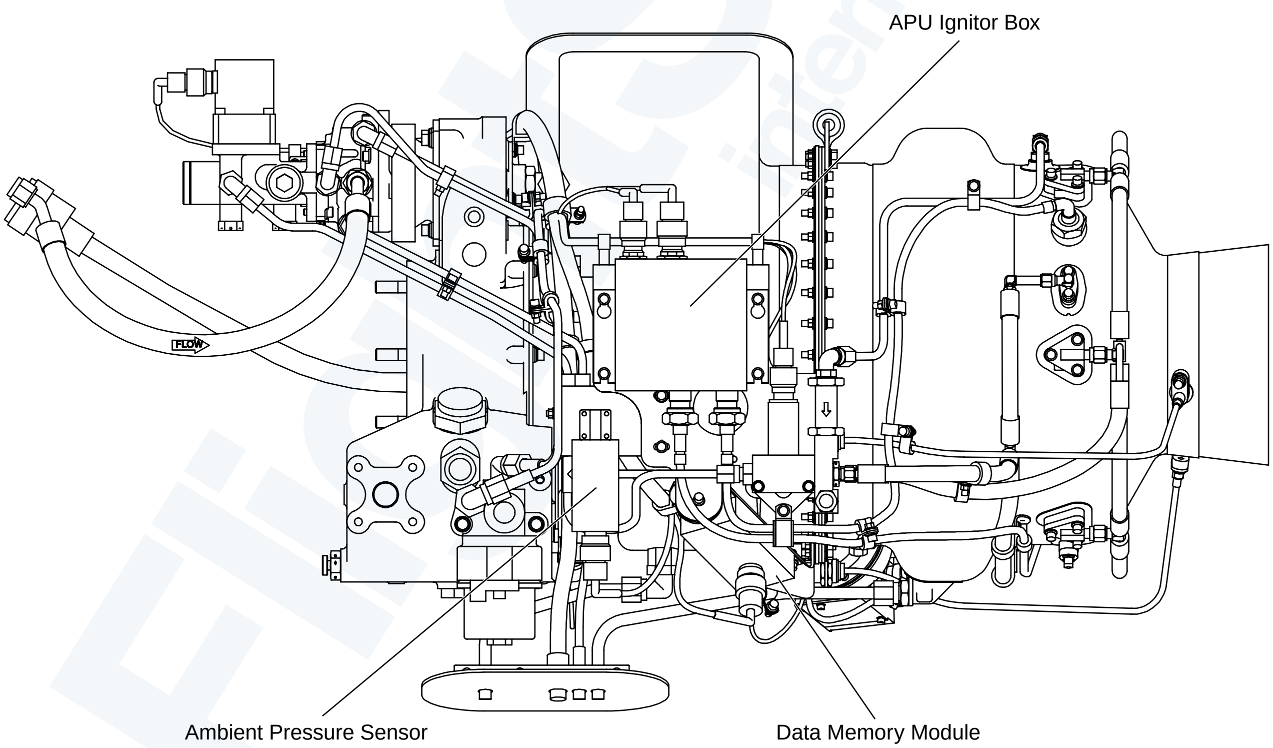


Figure 49-26. DMM and APS Sensor Locator

Speed Sensor

Refer to Figure 49-27. APU Sensors.

The sensor is installed on the air inlet housing, it is a single probe, dual coil device. The magnetic probe extends through the air inlet housing and aligns with the lobes on the quill shaft. The dual coils give redundant speed signals. The signals are transmitted to the FADEC and used for start sequencing and speed sensing (overspeed and underspeed). The two signals are averaged by FADEC to give the rated speed. If the difference between the two signals exceeds 5%, the FADEC will use the higher value. Failure of one signal will result in an advisory fault. Loss of both signals will cause an automatic APU shutdown.

Thermocouple

The two single-element chromel/alumel thermocouples are located on the APU exhaust flange and extend into the exhaust gas stream. The thermocouples are installed 90 degrees apart, at the five and eight o'clock positions and supply millivolt signals that are used by FADEC for exhaust gas temperature sensing. The FADEC averages the two signals to give the exhaust gas temperature. If the difference between the two thermocouples is greater than 93.3°C (200°F), the FADEC will use the higher value. An open or short failure of one thermocouple will result in an advisory message. Failure of both thermocouples will cause an APU shutdown.

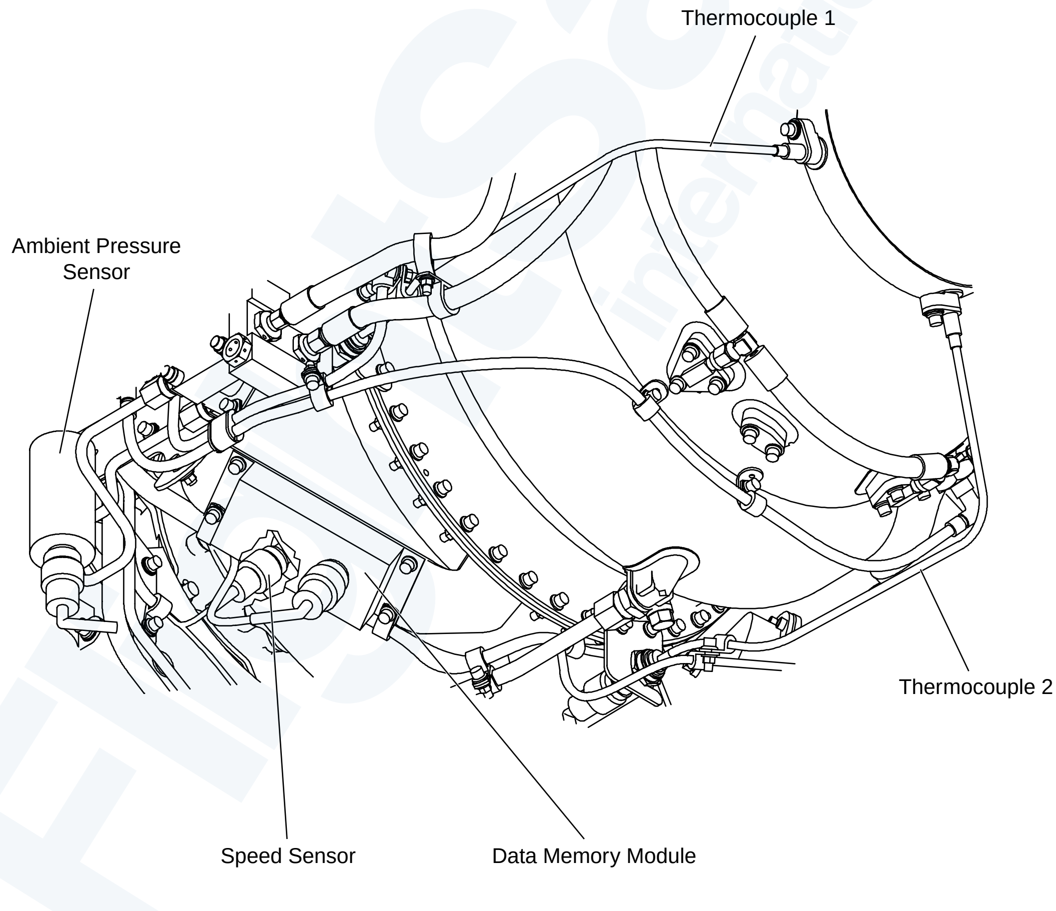


Figure 49-27. APU Sensors

49-40-00 IGNITION AND STARTING

INTRODUCTION

The APU ignition system is used to ignite the fuel while the starter provides the rotation of the APU.

GENERAL

The Ignition/Starting system includes the components that follow:

- APU CONTROL panel
- Starter/Generator
- Ignition exciter (1)
- Ignitor plugs (2).

COMPONENT DESCRIPTION

Starter/Generator

Refer to Figure 49-28. Starter/Generator Locator.

The Starter/Generator is an electromagnetic rotating machine that has two modes of operation.

Starter Mode

The unit converts 28 VDC power into mechanical torque that is used to drive the APU rotor for engine starting.

Generator Mode

The unit converts mechanical power from the gearbox into 28 VDC. The modes of operation are controlled by the FADEC and the Generator Control Unit (GCU).

The 28 VDC, 400 ‘A’ starter/generator is mounted on the forward face of the gearbox. The unit is mounted on a Quick Attach/Detach Adapter (QAD). The QAD incorporates the cooling air outlet. QAD is attached on six studs with self-locking nuts. A V-band clamp attaches the starter/generator to the QAD.

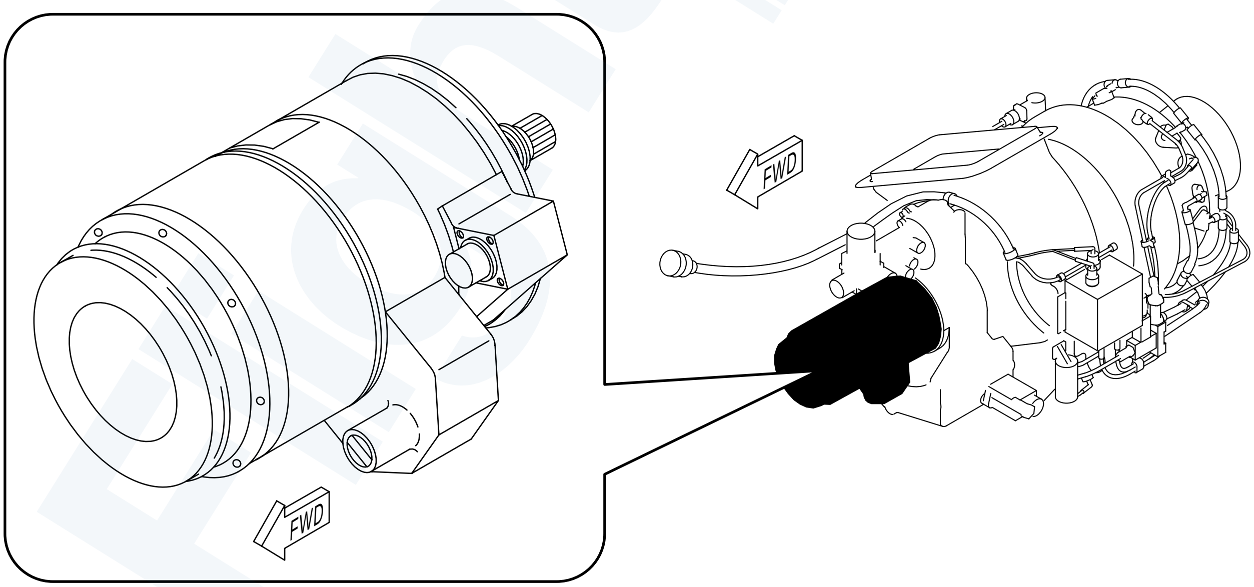


Figure 49-28. Starter/Generator Locator

The drive spline is lubricated by air oil mist from the gearbox. Any oil leakage picked up by the cooling airflow to blown overboard through the discharge duct. The starter/generator is an air cooled unit with an integral cooling fan. The inlet is titanium with a metal mesh screen. Each end of the composite duct is secured and sealed with a fireproof silicone composite boot and hose clamps. The outlet duct is bolted to the QAD. The outlet duct has a fireproof silicon seal at the point where it contacts the right access door. A bimetallic strip in the stator assembly provides a means for indicating a starter/generator overheat.

At 50% Ng the starter is de-energized. When the APU reaches 95% speed plus 3 seconds, the generator is available. The generator has the same generating capacity as the engine generators. The APU generators are not interchangeable with the engine units. The GCU regulates the generator output and controls the starter/generator contactor. The GCU is located under the cabin floor near the AC GCU's. Retain interchangeable with engine GCU's.

Ignition Exciter

Refer to Figure 49-29. APU Ignition System.

The ignition exciter box is installed on the left side of the engine. The exciter uses 28 VDC to give high voltage to the ignitor plugs. Operation of the exciter is controlled by the FADEC.

Ignitor Plugs

Two ignitor plugs are installed on the left and the right side of the engine at approximately the one and seven o'clock positions. The ignitor plugs are threaded into the engine combustor housing. The 10 millimeter gap type ignitors supply a high voltage intermittent spark that ignites the fuel-air mixture in the combustor.

Ignition Cable

The ignitors are connected to the ignition exciter by two separate shielded ignition cables. Wait a minimum of 5 minutes from the APU shutdown to release the high voltage from the ignition exciter.

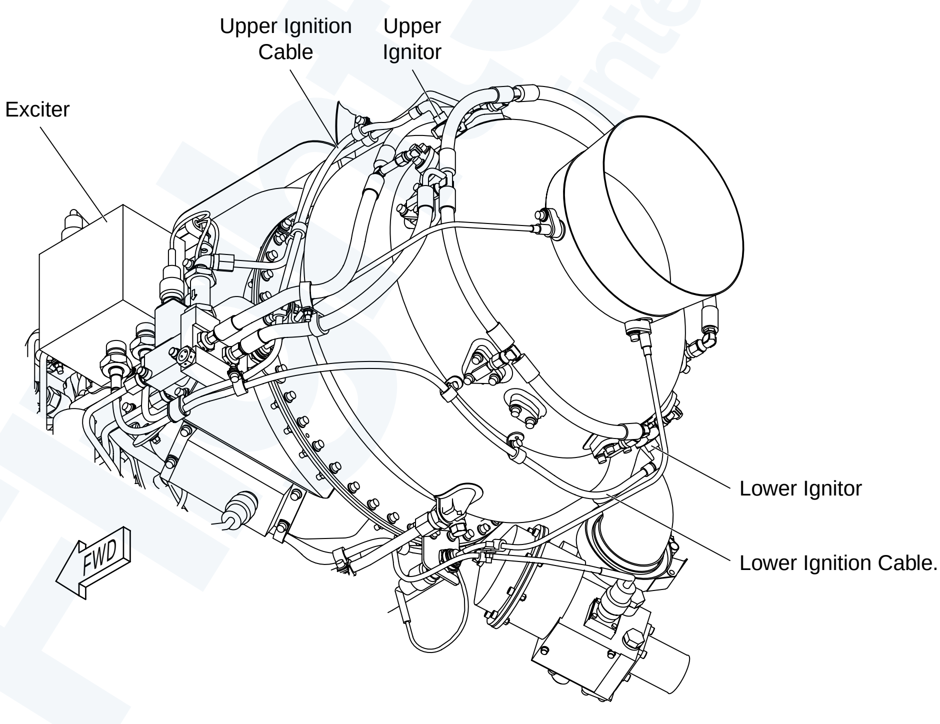


Figure 49-29. APU Ignition System

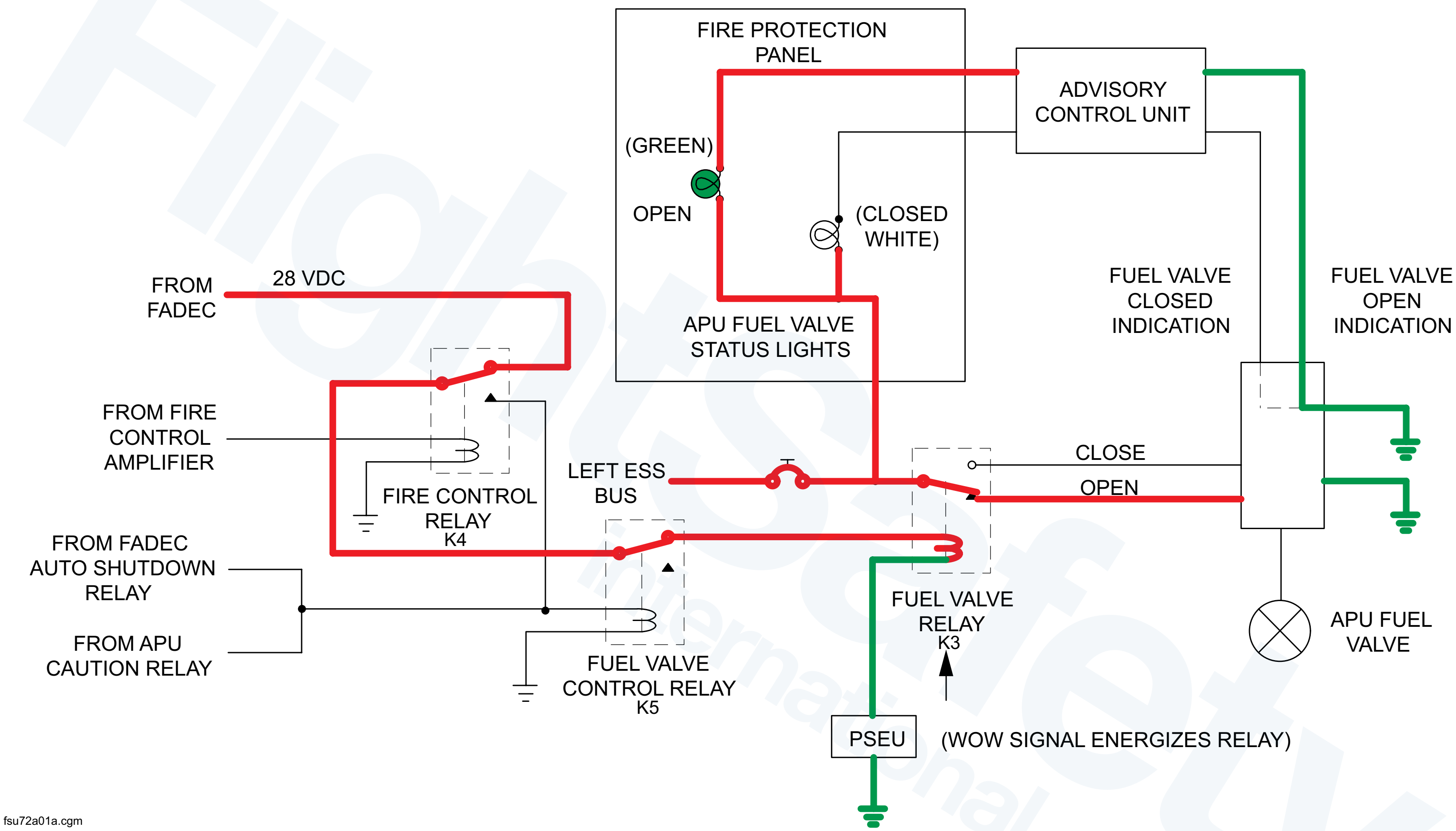


Figure 49-30. APU Fuel Valve Control Electrical Schematic

OPERATION

Refer to Figure 49-30. APU Fuel Valve Control Electrical Schematic.

The start and ignition sequence is automatically controlled by the APU FADEC.

Selection of the APU PWR switchlight, powers the FADEC which enables the APU functions. The FADEC will energize the K3 relay when a ground (WOW signal from the PSEU) is provided, this will power the APU Fuel Shutoff Valve to the Open position.

Pressing the START switchlight, the FADEC does the following:

- Sends a ‘Starter Operate’ signal to the GCU connector
- The APU GCU processes the starter operate signal and sends a 28 VDC Start Control signal to the DC contactor box to energize the relay K26, in the contactor box
- The Start Fuel valve is energized open at 3% Ng
- The Oil De-prime valve is energized closed
- The Igniters are energized. The capacitor ignition exciter receives 28 VDC and sends out intermittent high voltage to the igniter plugs. The gap type igniters give intermittent sparks which ignites the fuel-air mixture in the combustion chamber
- The Fuel Servo valve is modulated open
- The relay K26 sends 28 VDC from the right main bus to the APU Starter/Generator terminal. This causes the ‘starter’ to rotate the APU up to 50% Ng.

The APU 50% speed signal to the CGU causes the GCU to remove the 28 VDC from the relay K26 and de-energize the ‘starter’.

When the APU GCU senses the Ng is 95% plus 3 seconds, the generator function is available which is indicated as a WARN light on the APU CONTROL panel.

MAINTENANCE PRACTICES

The following are abbreviated descriptions of the maintenance practices and are intended for training purposes only. For a more detailed description of the practices, refer to the tasks in the Bombardier AMM PSM 1-84-2.

Igniter Plug Inspection

Visually inspect and replace if any of the following conditions are found:

- There are cracks or carbon build up on the ceramic parts
- There is erosion on the center electrode or the outer shell
- There is corrosion and damage on the barrel and threads.

TROUBLESHOOTING

Operational Test of the APU Igniter Plugs

- Disconnect the connectors, 4900-P305 and 4900-P306, to disable the start fuel solenoid
- Start the APU and make sure that there is a snapping sound of the igniters
- If there is no snapping sound, you must replace the igniters.

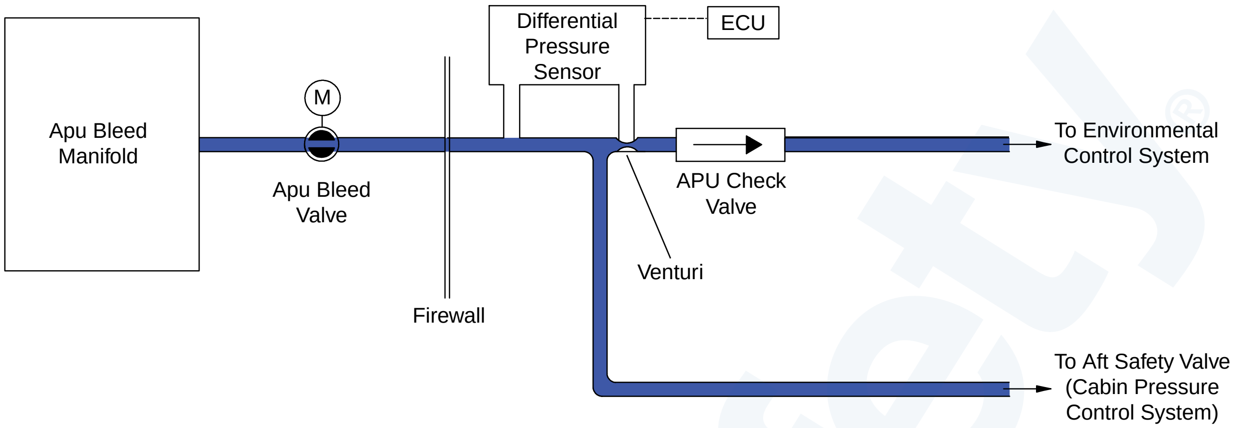


Figure 49-31. APU Bleed Air Schematic

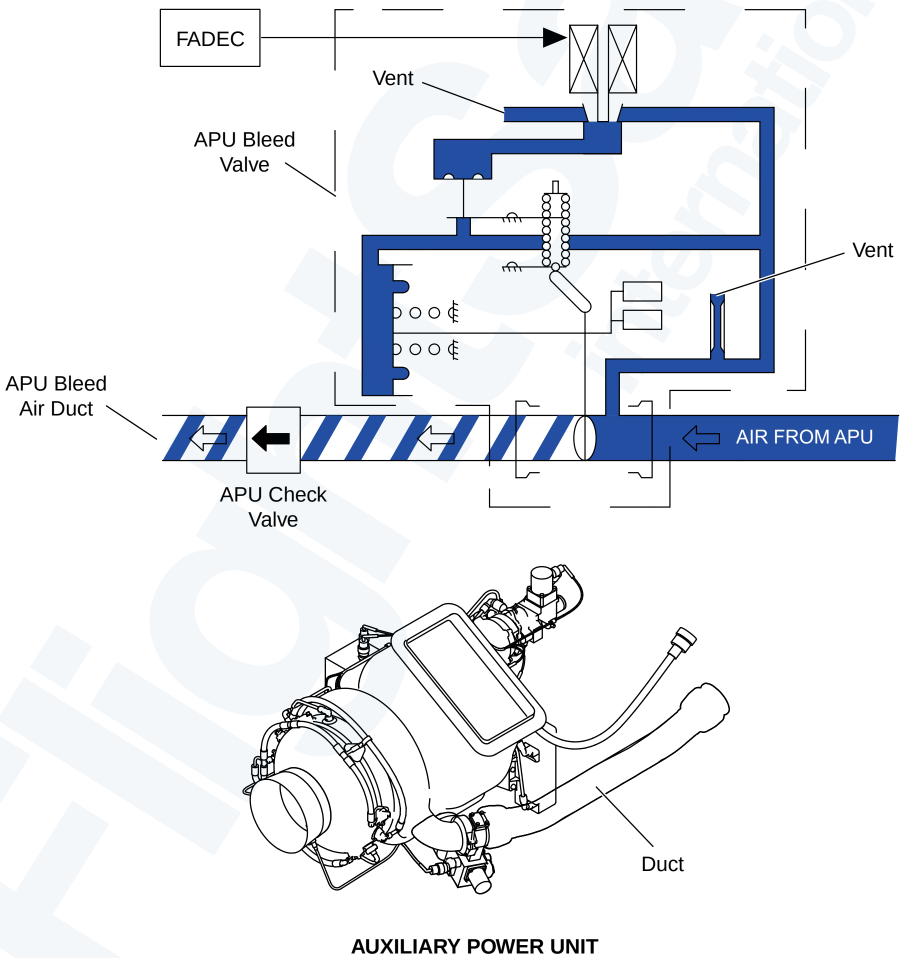


Figure 49-32. Bleed Valve and Duct

49-50-00 AIR

INTRODUCTION

The APU bleed air system supplies compressed air for combustion, the Environmental Control System (ECS) and other aircraft pneumatic systems.

GENERAL

Refer to Figure 49-31. APU Bleed Air Schematic.

The compressed air produced in the APU is used for its own operation and to supply air to the aircraft ECS.

COMPONENT DESCRIPTION

Auxiliary Power Unit (APU) Bleed Valve

Refer to Figure 49-32. Bleed Valve and Duct.

The bleed valve is an electro-pneumatic butterfly valve that is normally closed. The valve is located on the APU combustor housing bleed air duct and is controlled by the FADEC.

Bleed Air Duct

The bleed duct directs bleed air to the ECS. It is installed on the right side of the auxiliary power unit (APU).

CONTROLS AND INDICATIONS

When the BL AIR switchlight is selected OPEN on the APU CONTROL panel, the FADEC will open the bleed valve. The FADEC will control the modulation of the valve.

Selecting either engine BLEED 1 or 2 switches to ON will cause the FADEC to close the APU bleed valve.

When the BL AIR switchlight is selected, the OPEN indication on the APU control Panel will come on.

NOTES

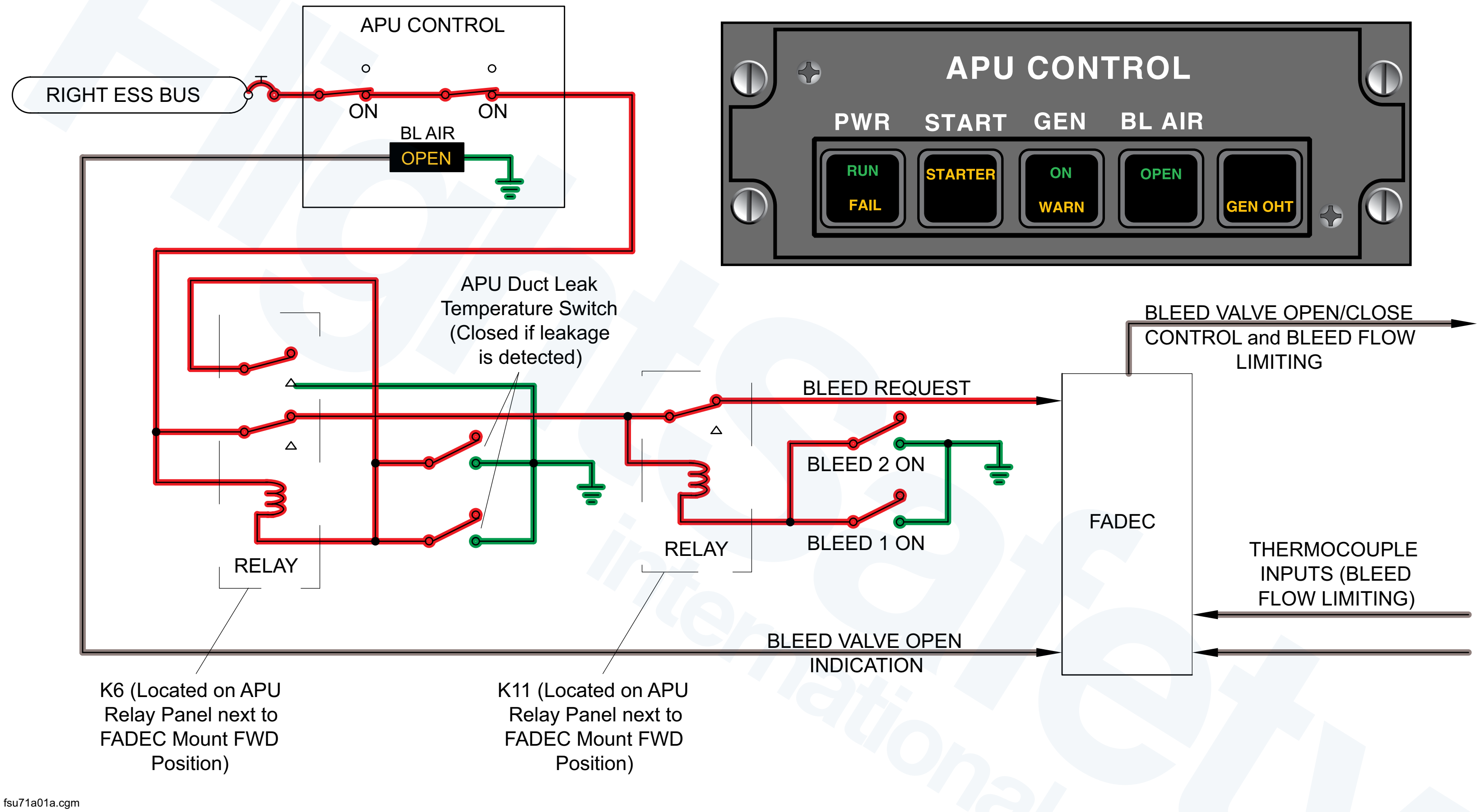


Figure 49-33. APU Control Panel and BL Air Switch Electrical Schematic

OPERATION

NOTES

Refer to Figure 49-33. APU Control Panel and BL Air Switch Electrical Schematic.

The normally closed bleed valve is controlled by the FADEC.

Selection of the BL AIR switchlight to OPEN on the APU CONTROL panel sends 28 VDC from the APU CONTROL panel to the APU relay panel. The 28 VDC goes through relay K11 and K6 to the FADEC as a 'Bleed Air Request'. This causes the FADEC to energize the Bleed air valve open.

If either Bleed Duct switch senses a duct air leak, relay K11 will close and remove the Bleed Air Request signal. Auxiliary contacts in K11 will latch the relay. To reset the system a selection of the Bleed Air switch to OFF then back to ON is required.

If either engine Bleed switch is selected to the on position, relay K6 will close and remove the Bleed Air Request signal.

The loss of the Bleed Air Request signal causes the FADEC to close the Bleed Air valve.

The FADEC also receives signals from the APU exhaust thermocouples for control of the Bleed Valve during bleed air delivery. As the APU EGT reaches 692°C the FADEC will modulate the Bleed Valve toward closed. This logic supplies bleed air without exceeding the APU EGT limits. Modulating bleed air flow in response to EGT limits gives the generator priority over bleed air.

The APU check valve in the bleed air duct is held open by the bleed air coming from the APU but stops the air from the PACKS to prevent reverse flow into the APU engine bleed air.

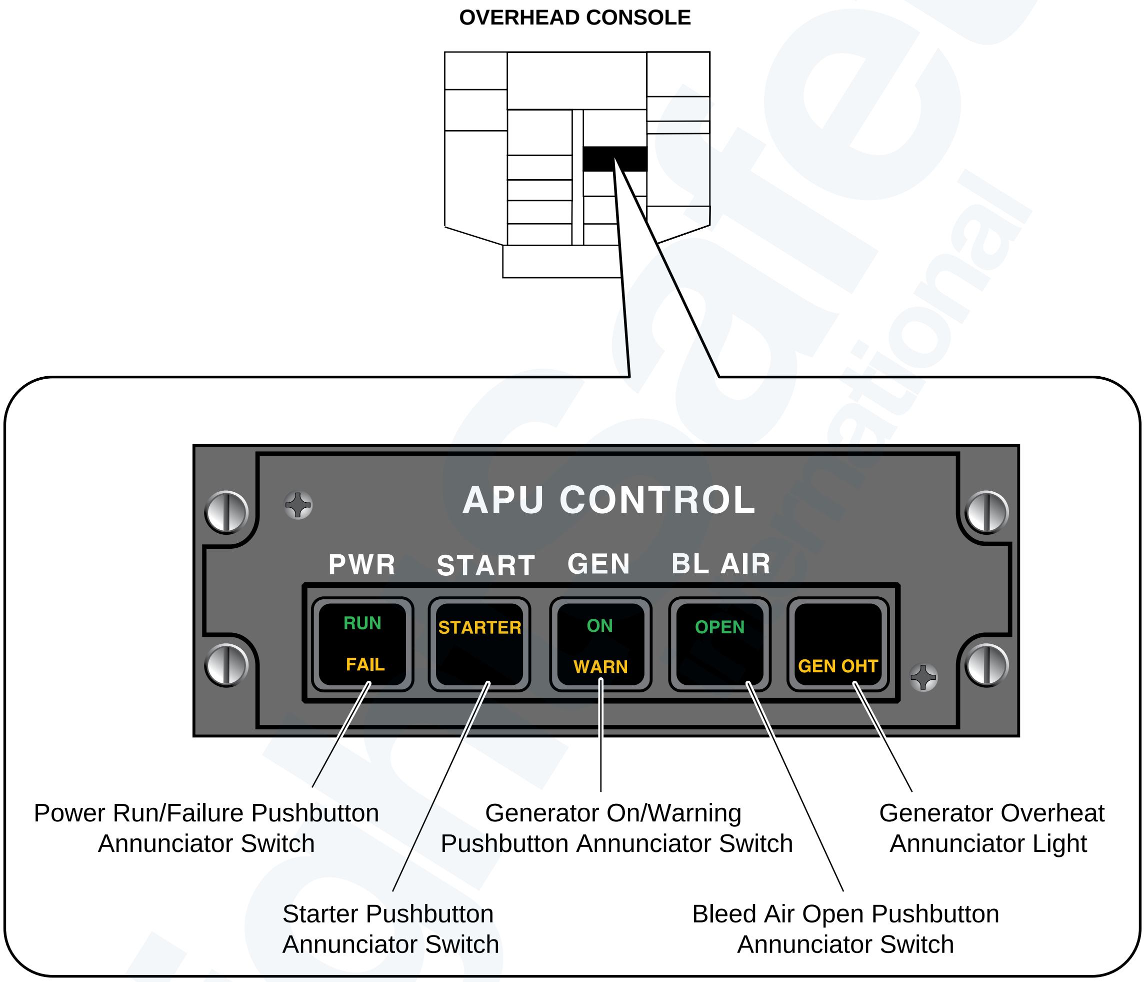


Figure 49-34. APU Control Panel

49-60-00 ENGINE CONTROLS

INTRODUCTION

Operation of the APU is controlled by the FADEC engine control system.

GENERAL

The APU is controlled by the APU control panel installed on the overhead console. The panel is supplied with 5 VDC background lighting when selected. The START switch is a momentary switch, the other switches are latched.

COMPONENT DESCRIPTION

The APU Control Panel

Refer to Figure 49-34. APU Control Panel.

PWR Switchlight

- Pushing the switchlight arms APU start circuits and opens the APU fuel valve
- A green RUN lighted segment will come on to show the APU is at operating speed
- An amber FAIL lighted segment will come on to show a failure is detected and the APU has been shutdown.

START Switchlight

- Pushing the STARTER switchlight will start the APU start cycle
- An amber STARTER lighted segment will come on to show the APU start sequence in progress.

GEN Switchlight

- Pushing the GEN switchlight causes the APU starter-generator to supply 28 VDC power to all the aircraft DC buses
- A green ON lighted segment will come on to show APU generator contactor has closed
- An amber WARN lighted segment will come on to show the APU starter generator contactor is open and DC power is available.

BL AIR Switchlight

- Pushing the switchlight will cause the APU bleed air valve to open
- A green OPEN lighted segment will come on to show the bleed valve is open.

GEN OHT Light

This light (amber) illuminated shows the APU starter-generator was in a overheat condition and APU automatically shut down.

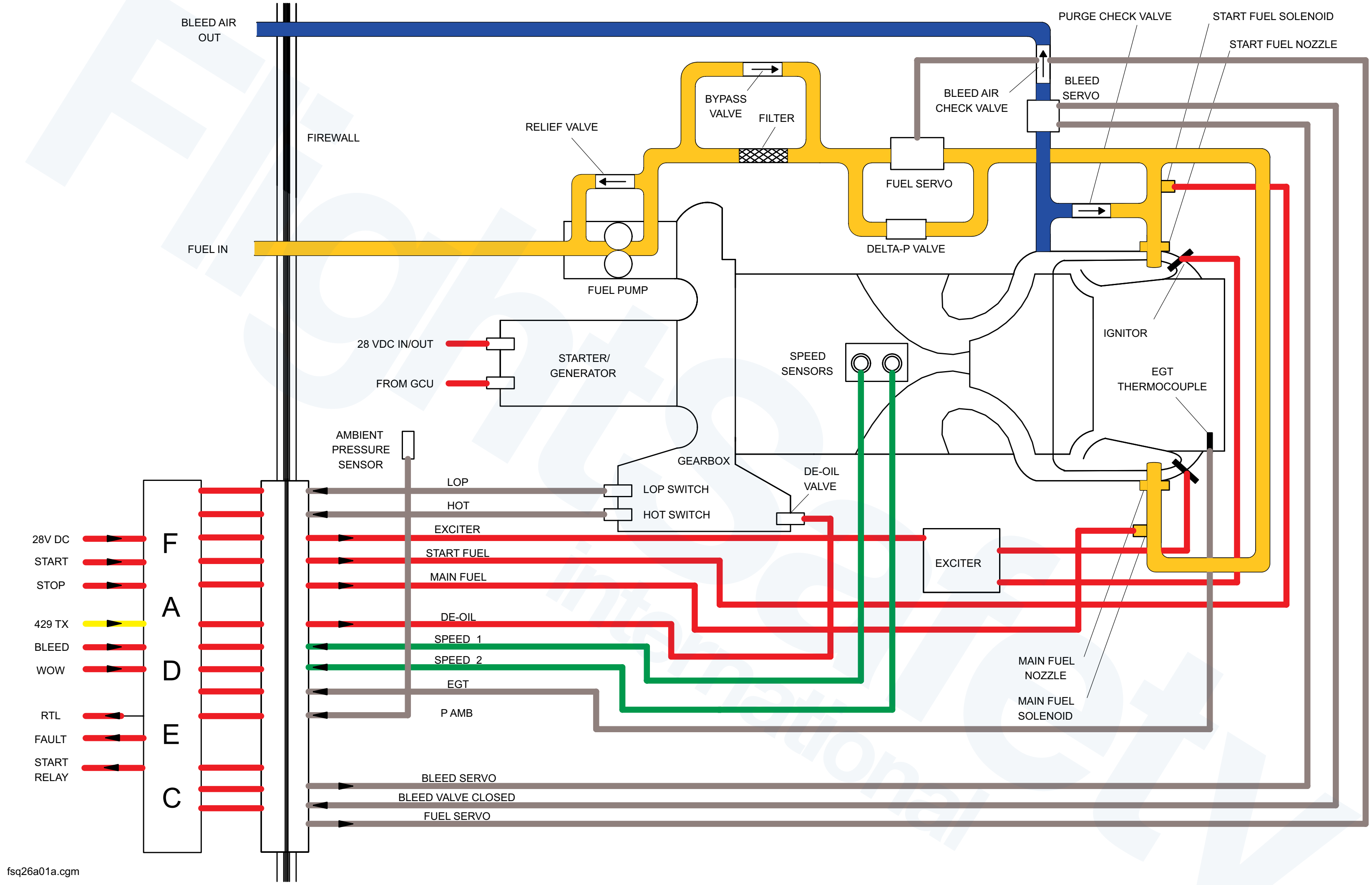


Figure 49-35. APU Control Schematic

The Full Authority Digital Electronic Controller (FADEC)

Refer to Figure 49-35. APU Control Schematic.

The FADEC is a microprocessor based controller, that can do control sequencing, condition monitoring, fault detection, fault isolation and protection of the APU system. The APU FADEC monitors the APU speed, oil temperature, oil pressure, and exhaust gas temperature. It controls the APU start up, fuel scheduling, manual and auto shutdown and overspeed protection. The FADEC transmits fault and maintenance data to the aircraft CDS through an ARINC 429 data link.

APU FADEC is located at the aft of the aircraft near the bottom center of the middle spar of the vertical stabilizer in the ECS compartment.

WARNING

OBEY THE FUEL SAFETY PRECAUTIONS WHEN YOU DO WORK ON THE FUEL SYSTEM OR A FUEL SYSTEM COMPONENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

GROUND THE AIRCRAFT AND THE WORKSTAND BEFORE YOU DO FUEL SYSTEM MAINTENANCE. AN ELECTROSTATIC SPARK CAN CAUSE AN EXPLOSION OR A FIRE.

CAUTION

DO NOT REMOVE THE DATA MEMORY MODULE (DMM) AND FULL AUTHORITY DIGITAL ELECTRONIC CONTROLLER (FADEC) AT THE SAME TIME. DO NOT REMOVE THE DMM OR FADEC BEFORE YOU PROGRAM THE REPLACEMENTS WITH THE CORRECT ENGINE DATA. THE REMOVAL OF THESE COMPONENTS BEFORE YOU PROGRAM THE NEW COMPONENTS WILL CAUSE THE LOSS OF THE ENGINE DATA.

THE DMM KEEPS THE SERIAL NUMBER DATA FOR THE APU. ALWAYS GIVE APPROXIMATELY 30 MINUTES TO LET THE TWO UNITS INTERFACE AFTER YOU CHANGE ONE OF THEM. DO NOT INTERCHANGE A DMM FROM ONE APU TO A DIFFERENT APU. THE DMM IS THE HISTORICAL RECORD OF THE APU. IF YOU INTERCHANGE A DMM WITH A DMM FROM A DIFFERENT APU, IT WILL CORRUPT THE APU DATA.

Start Sequence

Refer to Figure 49-36. Start Sequence.

The start sequence begins with the momentary start command signal from the start switch APU CONTROL panel. This opens the fuel servo valve.

The start fuel valve is energized open at 3% gas generator (Ng) speed which allows fuel flow to the two start nozzles and is ignited by the two ignitors. When 70% Ng speed is reached the main fuel valve is energized open which allows fuel flow to the six main fuel nozzles. A flow divider valve in the fuel manifold assembly opens when fuel pressure reaches 300 psid (2068.4 kPad). This makes sure proper fuel pressure is maintained in the start nozzles.

The start fuel valve is de-energized closed at 70% Ng. When the start valve closes, P3 bleed air opens the Purge Check valve which blows residual fuel from the start fuel manifold and nozzles.

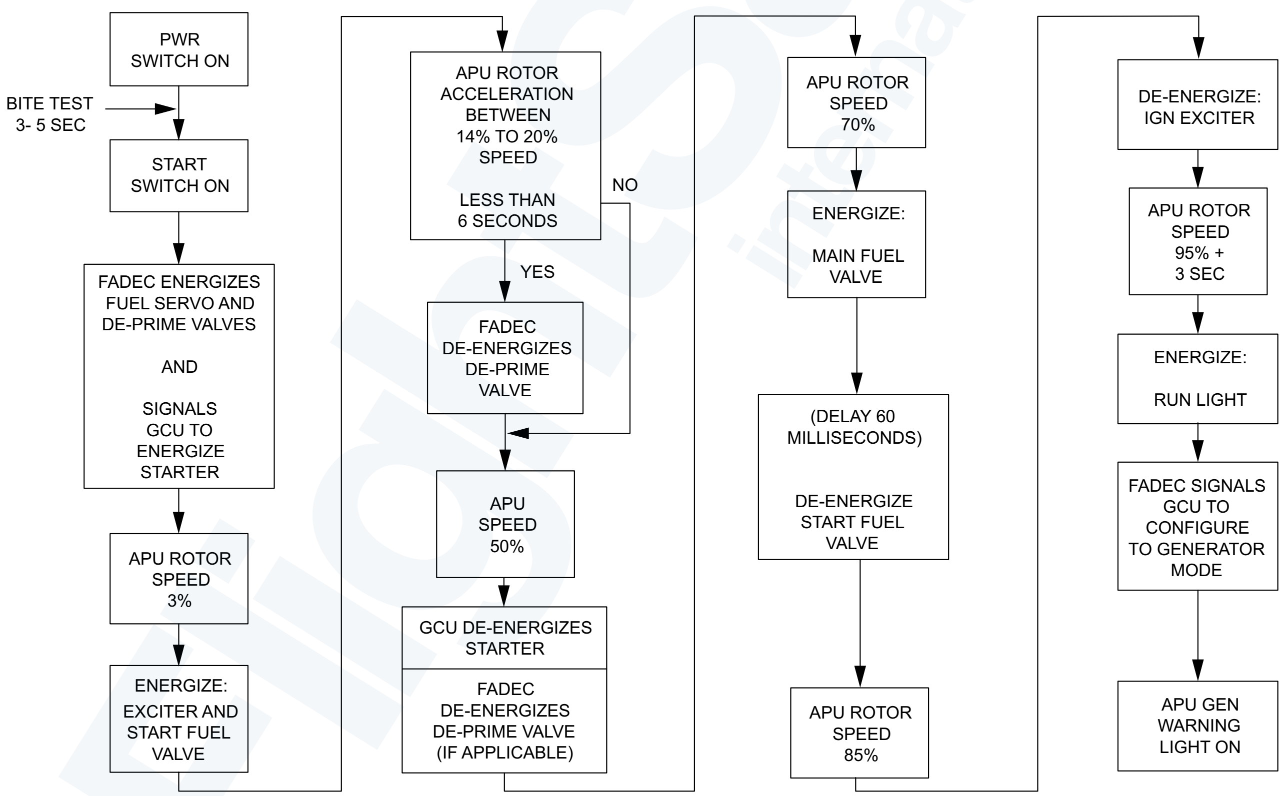


Figure 49-36. Start Sequence

Stop Sequence

Refer to Figure 49-37. Stop Sequence.

1. FADEC generated auto shutdown:

   Illogical switch input (example: APU ON switch pushed while either Weight Off Wheels (WOFW) signal from the PSEU is present)

   - Airborne operation
   - Low oil pressure
   - Both speed signals failed
   - Overspeed
   - Underspeed
   - Both EGT probes fail
   - High EGT
   - Main fuel valve failed (open, short, stuck)
   - High oil temperature
   - FADEC internal timer fail.

2. Aircraft generated auto shutdown:
   - APU fire detection system
   - APU fire bottle low pressure
   - Generator overheat
   - #1 WOW indicating air mode
   - #2 WOW indicating air mode.

At any time that a major failure is detected by the FADEC, the APU is automatically shutdown. The 'fault' signal closes the fuel valves and illuminates the APU Caution light in the flight compartment, the WARN annunciation on the APU control panel, and extinguishes the RUN light.

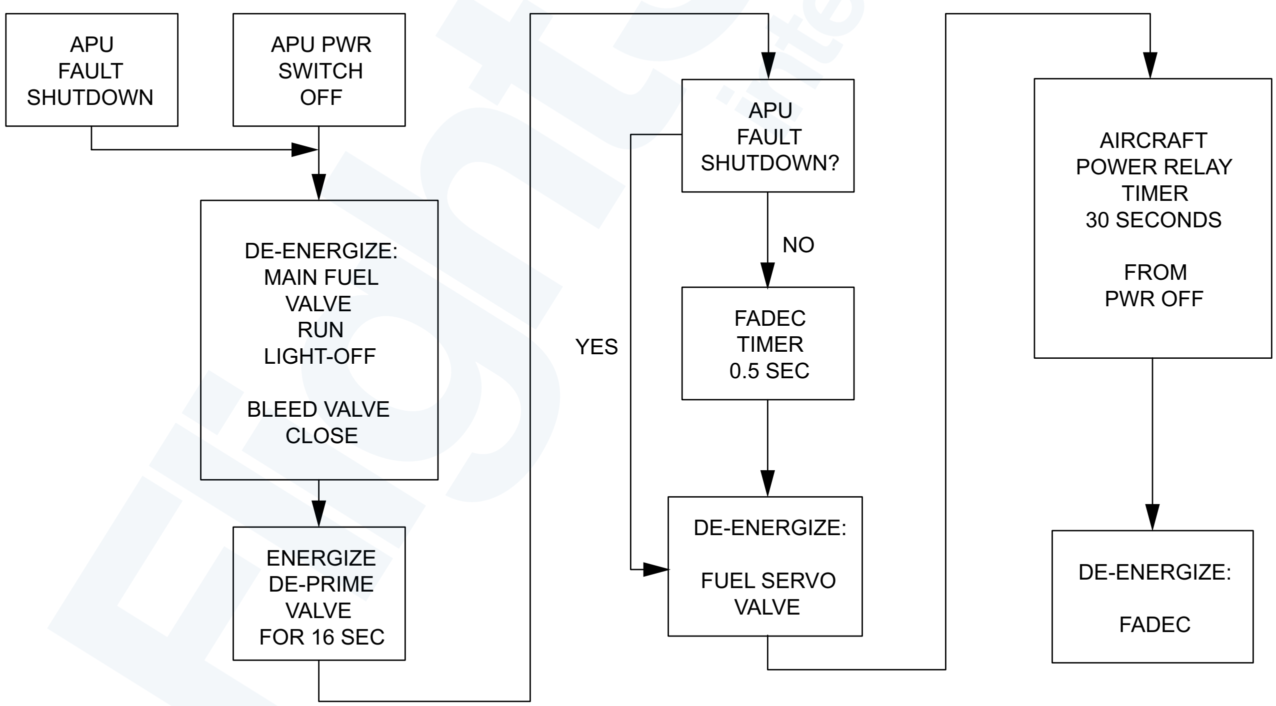


Figure 49-37. Stop Sequence

MAINTENANCE PRACTICES

The following is an abbreviated description of the maintenance practice and is intended for training purposes only.

For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

APU Starting

Before starting the APU make sure the left fuel tank has a minimum of 250 lb. (113 kg) of fuel. Do a test of the advisory lights.

Select the APU power switchlight to ON.

Make sure the APU FUEL VALVE OPEN light comes on and CLOSED light goes off. Also make sure the APU does a successful FADEC BIT check.

Do an APU fire test.

Obey the start limits of 1 minute ON 5 minutes OFF and 1 minute ON 30 mintues OFF.

CAUTION

IF THE APU DOES NOT START SUCCESSFULLY ON TWO ATTEMPTS, OBEY THE START ON AND OFF LIMITS BEFORE SUBSEQUENT START ATTEMPTS. MAINTENACNE CORRECTIVE ACTION IS NECESSARY AFTER THE SECOND UNSUCCESSFUL START, YOU CAN CAUSE DAMAGE TO THE APU.

Limitations:

- Ambient Running Temperature: -54°C to +50°C
- Starting Temperature: -35°C to 50°C (< -35°C pre-heat is required)
- Max. Operating +21°C with winter (louvered) cover installed.

To start the APU push the START button on APU CONTROL panel. The STARTER light will stay on until the starter disengages (50%Ng). The RUN light will come on when APU reaches the ready to load condition.

49-70-00 APU INDICATING

INTRODUCTION

The Indicating System displays the APU operating parameters and conditions sensed by the FADEC.

GENERAL

Refer to Figure 49-38. APU Fire Protection.

The Indicating System has the components that follow:

- APU CONTROL PANEL
- MFD ELECTRICAL PAGE
- APU FIRE PROTECTION PANEL
- CAUTION AND WARNING PANEL
- CDS
- ARCDU 1 and 2.

APU Fire Protection Panel

1. VALVE CLOSED Light is on when the APU fuel shut-off valve is closed.
2. FUEL OPEN Light is on when the APU fuel shut-off valve is open.

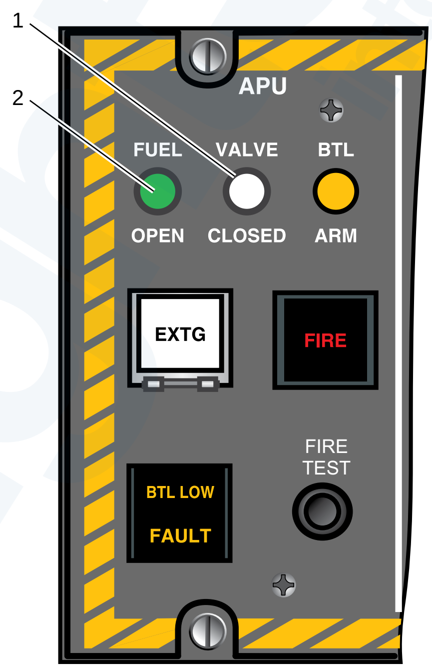


Figure 49-38. APU Fire Protection

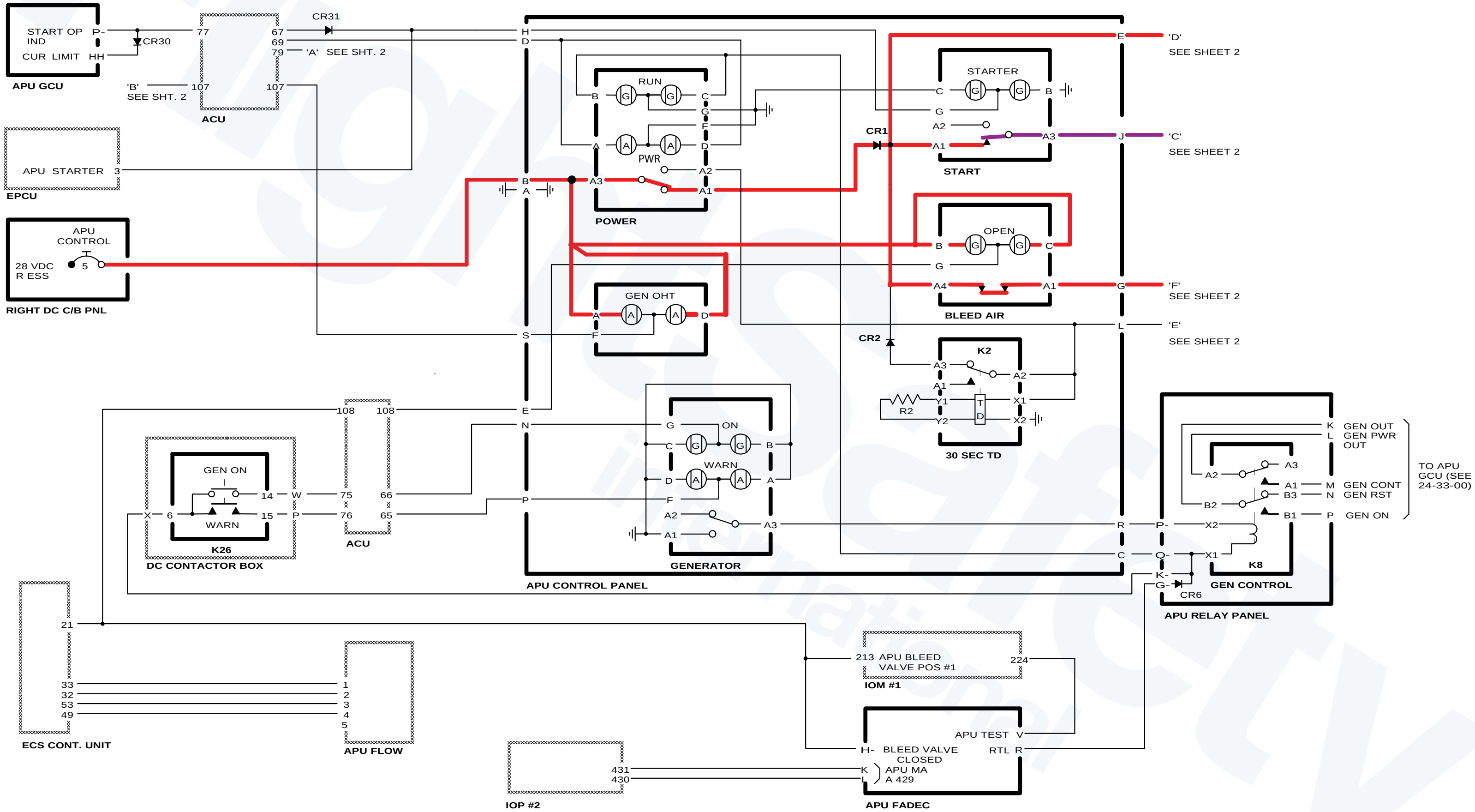


Figure 49-39. APU Control (Sheet 1 of 2)

OPERATION

Refer to:

- Figure 49-39. APU Control (Sheet 1 of 2).
- Figure 49-40. APU Control (Sheet 2 of 2).

The APU FADEC provides control and indication and records APU data.

PWR Switchlight is Pushed

- This arms the APU start circuits and opens the APU fuel valve
- FUEL OPEN green light comes on.

START Switchlight is Pushed

- This energizes APU start sequence
- STARTER amber light comes on
- At 50% Ng the STARTER light will go off
- After APU reaches 95% for 3 seconds, RUN green light comes on
- APU generator is in standby
- WARN amber light comes on.

GEN WARN Switchlight is Pushed

- The APU generator sends 28 VDC to the right main DC bus
- ON green light comes on.

BL AIR Switchlight is Pushed

- APU bleed air valve opens
- OPEN green light comes on.

FAIL Amber Light Comes On

- An APU failure is detected and the FADEC shuts down the APU automatically
- APU bleed air valve is closed
- APU fuel valve is closed
- VALVE CLOSED white light comes on
- APU amber caution light comes on.

(Refer to the FIM 49-70-00-02 Indicating - Maintenance Practices)

GEN OHT Amber Light Comes On

- The APU DC starter/generator temperature is more than 166°C (330°F)
- An APU failure is detected and the FADEC shuts down the APU automatically
- APU bleed air valve is closed
- APU fuel valve is closed
- VALVE CLOSED white light comes on
- APU amber caution light comes on.

During start initiation, the FADEC uses EGT thermocouple for reference temperature to sense a rise in EGT before the start sequence can continue. The FADEC uses the speed sensor data for fuel to sequence the fuel valves and ignition.

The Speed sensor dual coils give redundant speed signals to the FADEC and for the start sequence, over-speed and under-speed sensing. The FADEC creates an average of the two signals to create the rated speed.

If the two signals difference exceeds 5%, the FADEC will use the higher value.

One signal fail will cause an advisory message found in the CDS.

Both signals fail will cause the FADEC to automatically shutdown the APU.

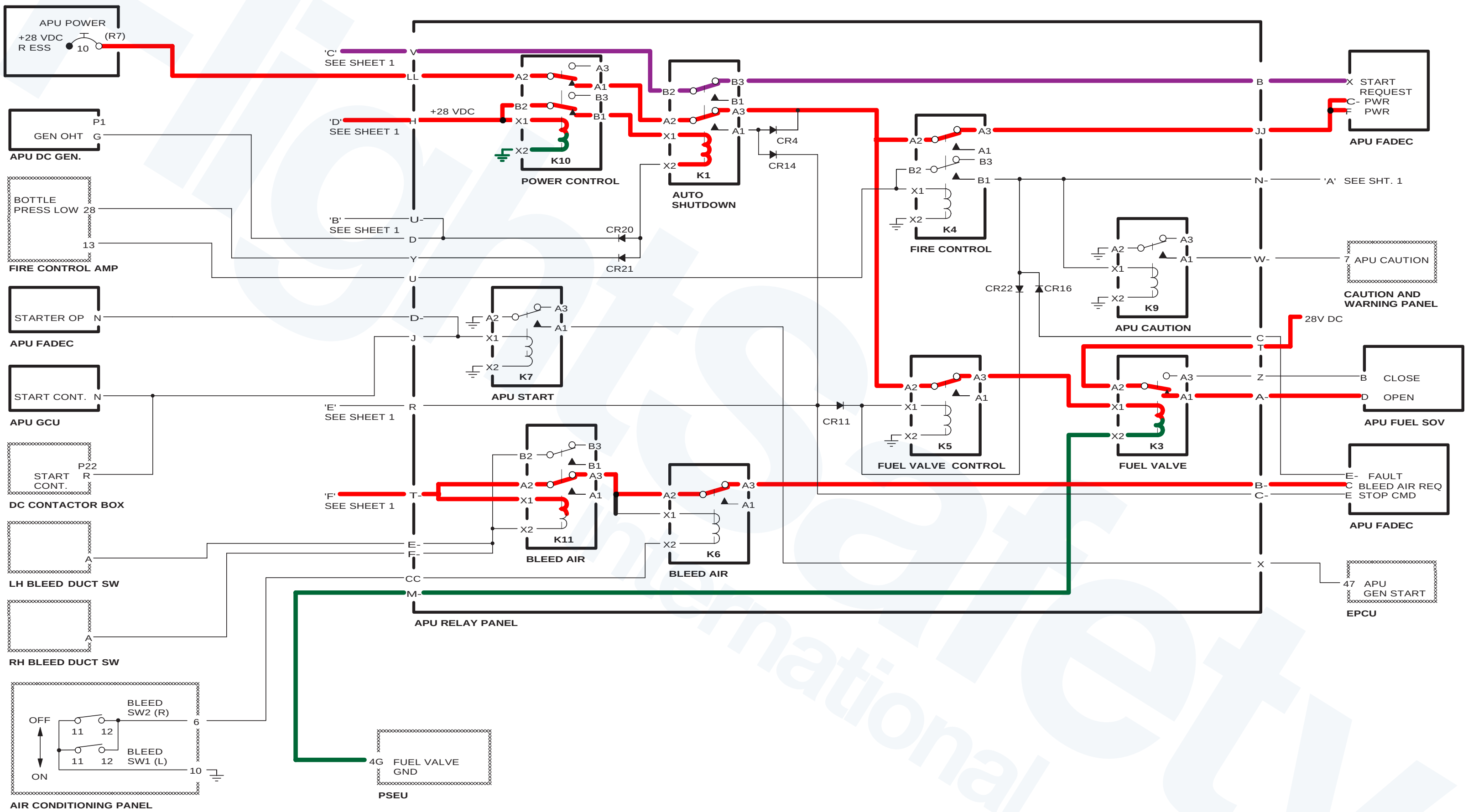


Figure 49-40. APU Control (Sheet 2 of 2)

Two single element thermocouples supply a millivolt signal to the FADEC for exhaust gas temperature sensing. The FADEC averages the two signals for EGT sensing.

If the two signals difference exceeds 200°F (93.3 °C), the FADEC will use the higher value.

One signal fail will cause an advisory message found in the CDS.

Both signals fail will cause the FADEC to automatically shutdown the APU.

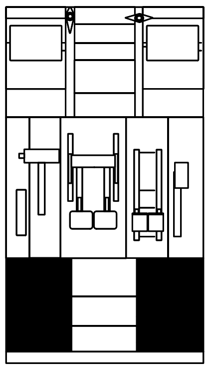
The High Oil Temperature normally open switch closes if the oil temperature exceeds 275°F (135°C).

A closed switch will cause FADEC to automatically shutdown the APU.

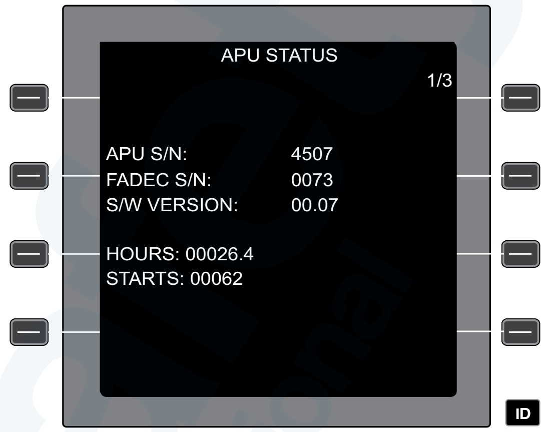
The Low Oil Pressure normally closed switch senses the oil pressure downstream of the oil filter and opens when oil pressure exceeds 35 psig.

An open switch prior to start (failed open), or a closed switch during operation will cause the FADEC to automatically shut down the APU.

NOTES



CENTER CONSOLE

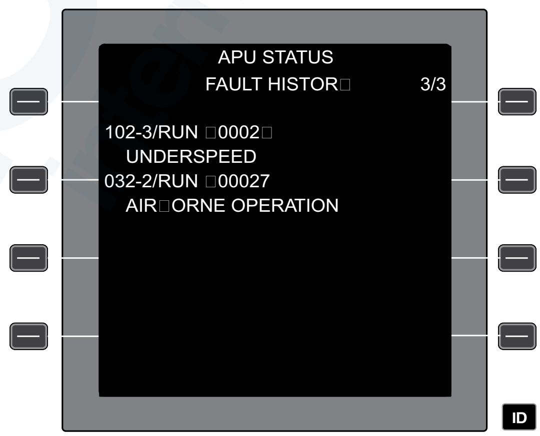


APU STATUS Page 1 of 3

Step #1 : Select Next

APU STATUS Page 2 of 3

Step #2 : Select Next ⟶
Or Select Prev ⟵



APU STATUS Page 3 of 3

Step #3 : Select Next ⟶ To Continue
Or Select Prev ⟵ To Go Back A Screen
Or Dim/rtn To Return To The Cds Menu

Figure 49-41. CDS Pages

FAULT DIAGNOSTICS

There are two methods of retrieving faults from the APU FADEC. One is through the CDS the other is by direct connection to the FADEC using a data transfer cable to a laptop equipped with software that will analysis and display the Data.

Centralized Diagnostic System

The Centralized Diagnostic System (CDS) allows access to electronic APU fault detection and fault storage data. The CDS displays the fault codes operational testing and troubleshooting.

Refer to Figure 49-41. CDS Pages.

Central Diagnostic System Unit

The CDS will display the APU status data over three pages. Page one will display APU serial number, FADEC serial number, software version, Hours of operation and number of starts. Page two will display current faults if faults are present. Page three will display the fault history of the APU.

Downloading FADEC Data

The APU PWR switch must be in the ON position with the APU not operating. The switch must stay ON for the duration of the Smart Terminal inquiry. Connect the Smart Terminal cable between the laptop and the J3 connector on the FADEC. Start the Smart Terminal to download the parametric data for troubleshooting. Follow the on-screen instructions for operation of the troubleshooter.

Audio and Radio Control and Display Units (ARCDU)

The ARCDU 1 or ARCDU 2 can display the CDS menus which includes the APU fault information sent by the FADEC to the CDS unit.

APU Status Pages

There are three types of APU STATUS pages, titled and numbered as follows:

- APU STATUS - 1/3
- APU STATUS PRESENT FAULTS 2/3
- APU STATUS FAULT HISTORY 3/3.

The CDS APU STATUS page 1/3 provides the following information:

- APU Serial Number
- FADEC Serial Number
- Software Version Number
- APU Hours
- APU Starts.

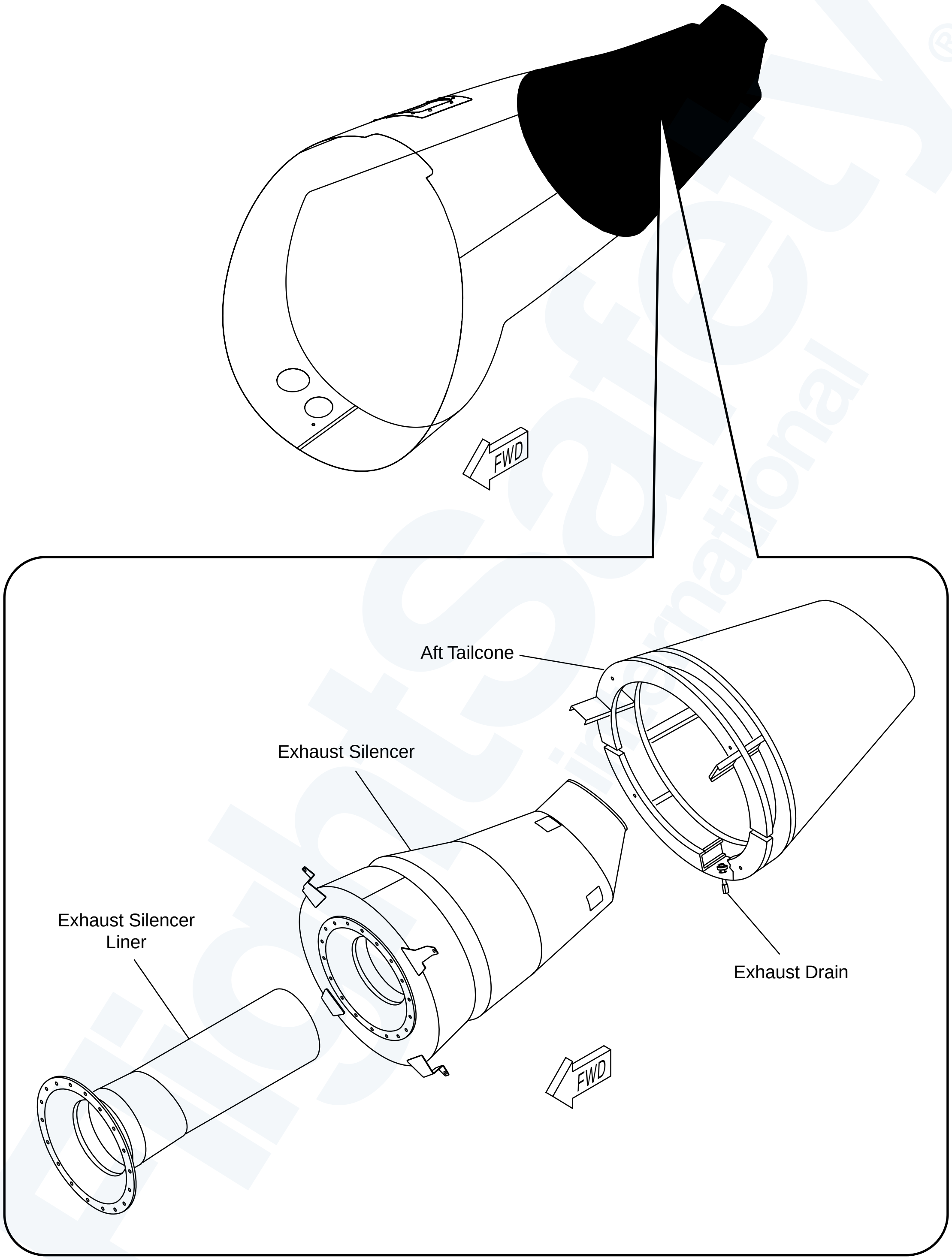


Figure 49-42. Exhaust Drain

49-80-00 APU EXHAUST

NOTES

INTRODUCTION

The APU exhaust system vents the exhaust gases from the APU to the environment and ventilates the APU compartment.

COMPONENT DESCRIPTION

APU Exhaust Silencer

The exhaust silencer is installed in the aft tailcone.

APU Exhaust Liner

The exhaust silencer liner is a felt metal liner installed in the auxiliary power unit (APU) exhaust silencer. It directs the exhaust gases away from the APU.

Gearbox Cooling Duct

The APU gearbox cooling duct is installed in the tailcone, forward of the Clamshell doors. It is a 4 inch duct protected by a stainless steel mesh screen. It directs cooling air towards the front of the oil sump to assist in oil cooling.

Exhaust Drain

Refer to Figure 49-42. Exhaust Drain.

The exhaust drain is a stainless steel tube threaded into the exhaust silencer to drain moisture.

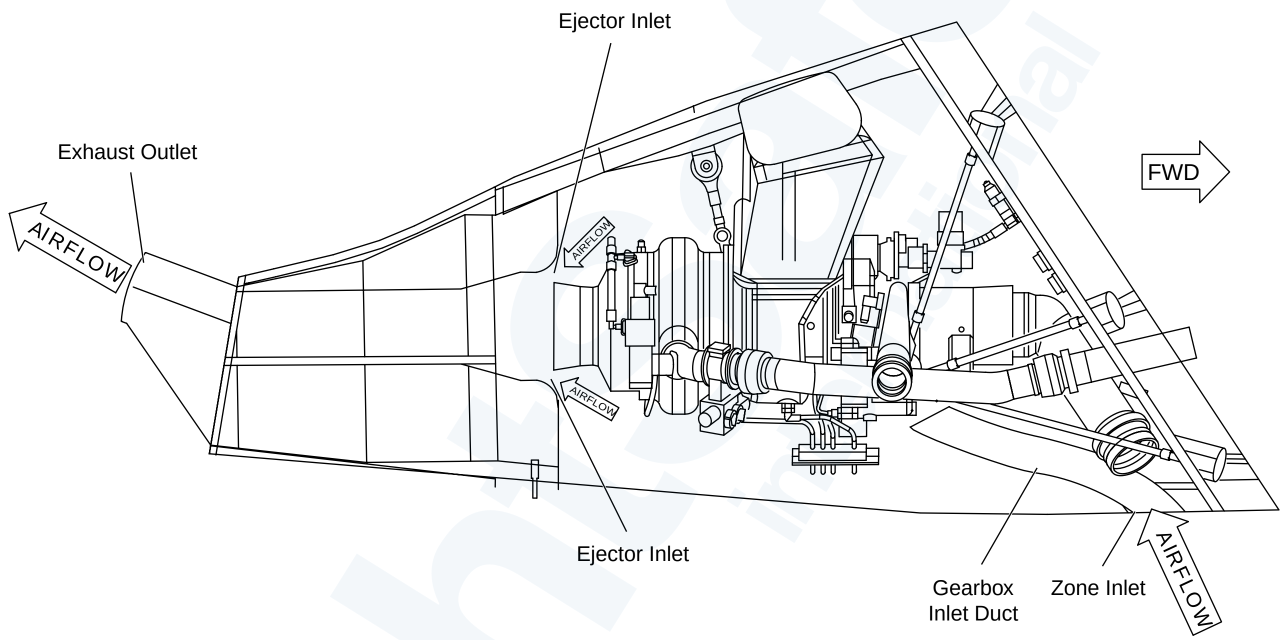


Figure 49-43. APU Exhaust

OPERATION

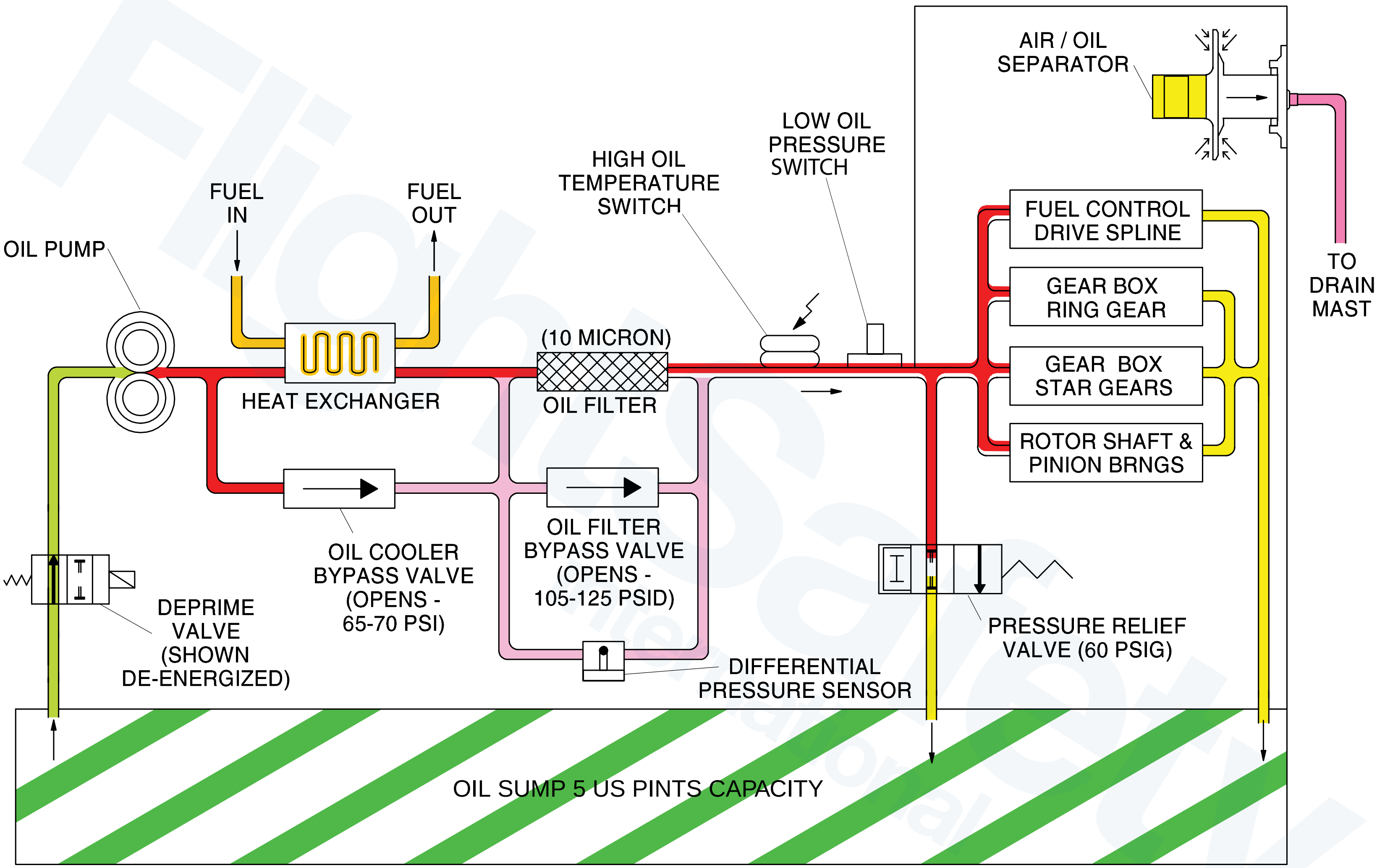
Refer to Figure 49-43. APU Exhaust.

The APU exhaust system vents the exhaust gases from the APU and ventilates the APU compartment. The APU and compartment are kept cool by the exhaust flow.

The APU exhaust nozzle and the silenced exhaust duct act as an ejector system.

Ambient air enters through the stainless steel mesh screen inlet just forward of the APU access doors. From the inlet the air is ducted to the front face of the oil sump for cooling. The air flows around the APU to be sucked into the ejector gas flow. The ejector pump uses the dynamic energy of the jet exhaust from the APU nozzle to suck compartment ambient air into the annular gap between the APU nozzle and the exhaust duct.

NOTES



fss89a01a.cgm

Figure 49-44. Oil System Schematic

49-90-00 OIL

INTRODUCTION

The APU oil system supplies pressurized oil to the gearbox and the engine rotor assembly to lubricate and cool the gears and the bearings.

GENERAL

Refer to Figure 49-44. Oil System Schematic.

The oil system contains the components that follow:

- Gear type pump
- De-prime valve
- Fuel/oil cooler
- Fuel/oil cooler bypass valve
- Filter
- Pressure relief valve
- High temperature switch
- Low pressure switch
- Oil/air separator
- Carbon seal drain
- Labyrinth seal
- Oil filler port
- Sight glass
- Magnetic chip detector/drain plug.

NOTES

COMPONENT DESCRIPTION

Oil Pump

The positive displacement gear type oil pump is located inside the gearbox. It supplies oil flow to the APU system.

Deprime Valve

Refer to Figure 49-45. Oil System Components.

The deprime valve is a solenoid operated valve that is located on the lower, right side of the gearbox. The valve is energized closed by the FADEC to prevent the oil pump from supplying oil during engine startup and shutdown. This reduces the load during cranking to improve acceleration. Above 50% to allow engine speed the deprime valve is de-energized to allow oil flow. During engine shutdown the deprime valve is energized for 16 seconds to stop oil pump output.

Fuel/Oil Cooler

The Fuel/Oil cooler is located on the lower, right side of the gearbox. The cooler uses fuel to cool the oil.

Fuel/Oil Cooler Bypass Valve

The valve is located in the gearbox. If the oil flow is restricted through the cooler, the cooler bypass valve opens at 448 to 483 kPag (65 to 70 psig).

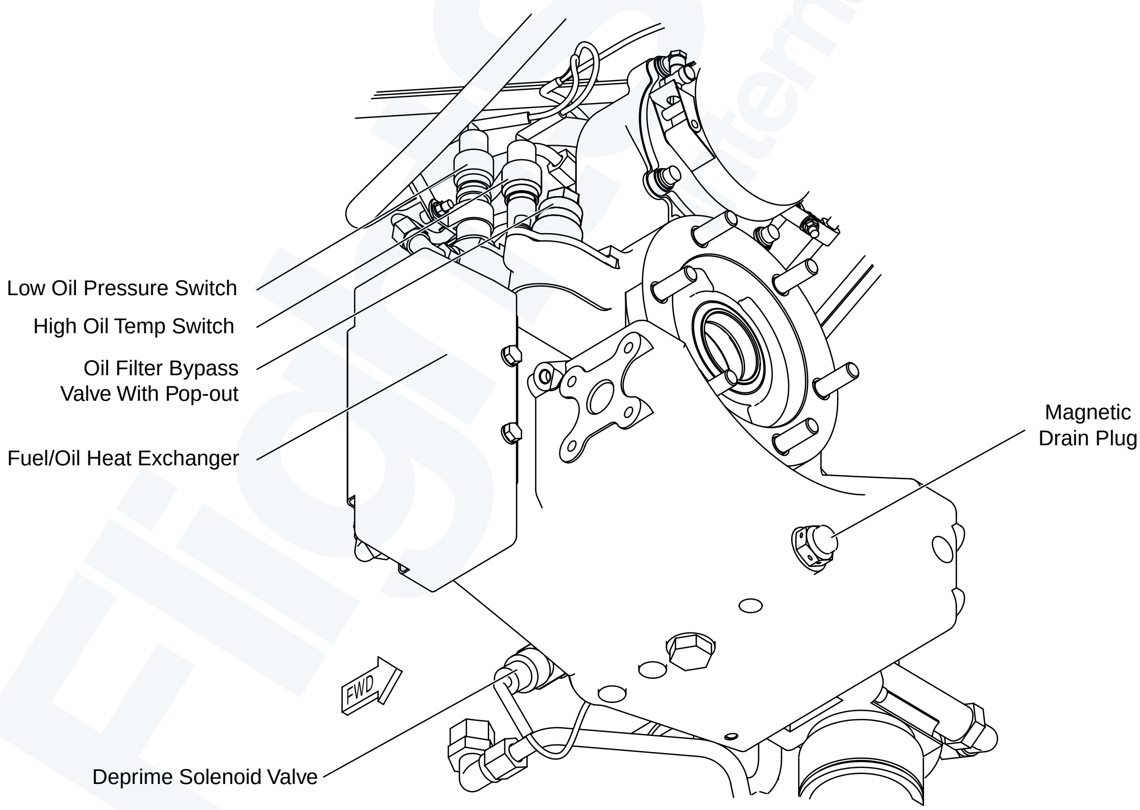


Figure 49-45. Oil System Components

Oil Filter

The oil filter is a disposable 10 micron filter. The oil filter bypass valve assembly includes a Pop-UP bypass indicator. The indicator is visible when differential pressure across the filter is greater than 60 to 80 psid (414 to 552 kPad).

Oil Pressure Relief Valve

The oil pressure relief valve, located between the oil filter and the gearbox oil jets, maintains the system oil pressure of 414 kPag (60 psig). The valve is a non-adjustable and non-replaceable component.

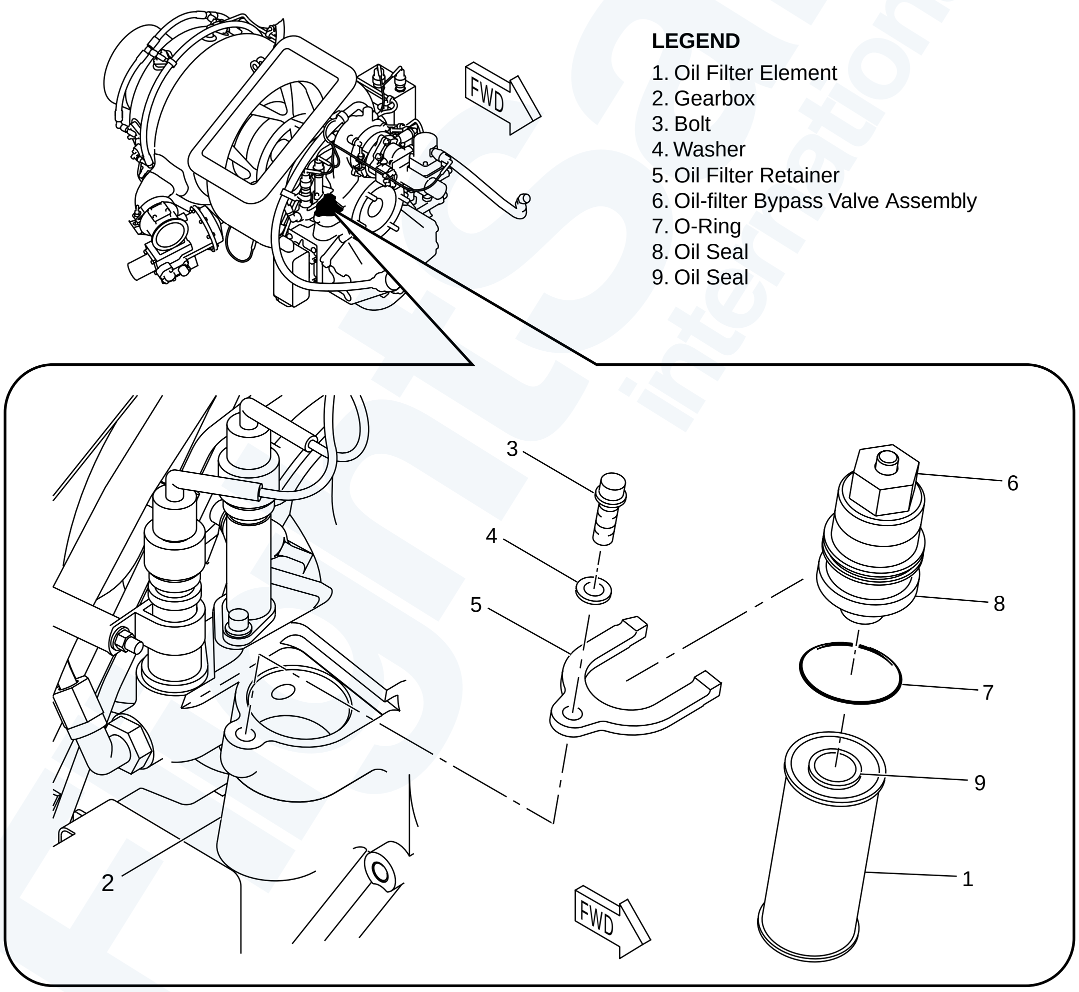


Figure 49-46. Oil Filter

High Oil Temperature Switch

The high oil temperature switch is a normally open switch and closes when the oil temperature is more than 275 ±5 F (135 ±3 C). A closed switch causes the APU to shutdown.

Low Oil Pressure Switch

The Low Oil Pressure Switch senses oil pressure downstream of the oil filter. The open switch opens when oil pressure is 35 ±5 psig (241 ±34.5 kPag) and remains open during normal operation of the APU. If low Oil pressure is sensed when Ng is 100%, the switch will close causing the APU to shut down.

Air/Oil Separator

The oil/air separator is part of the fuel control gear and separates the oil from the air inside the gearbox. The oil returns by gravity to the oil sump and separated air is vented overboard through the drain mast vent.

Refer to Figure 49-47. Carbon and Labyrinth Seals.

Carbon Seal

The carbon face seal prevents oil in the bearing area from entering the compressor.

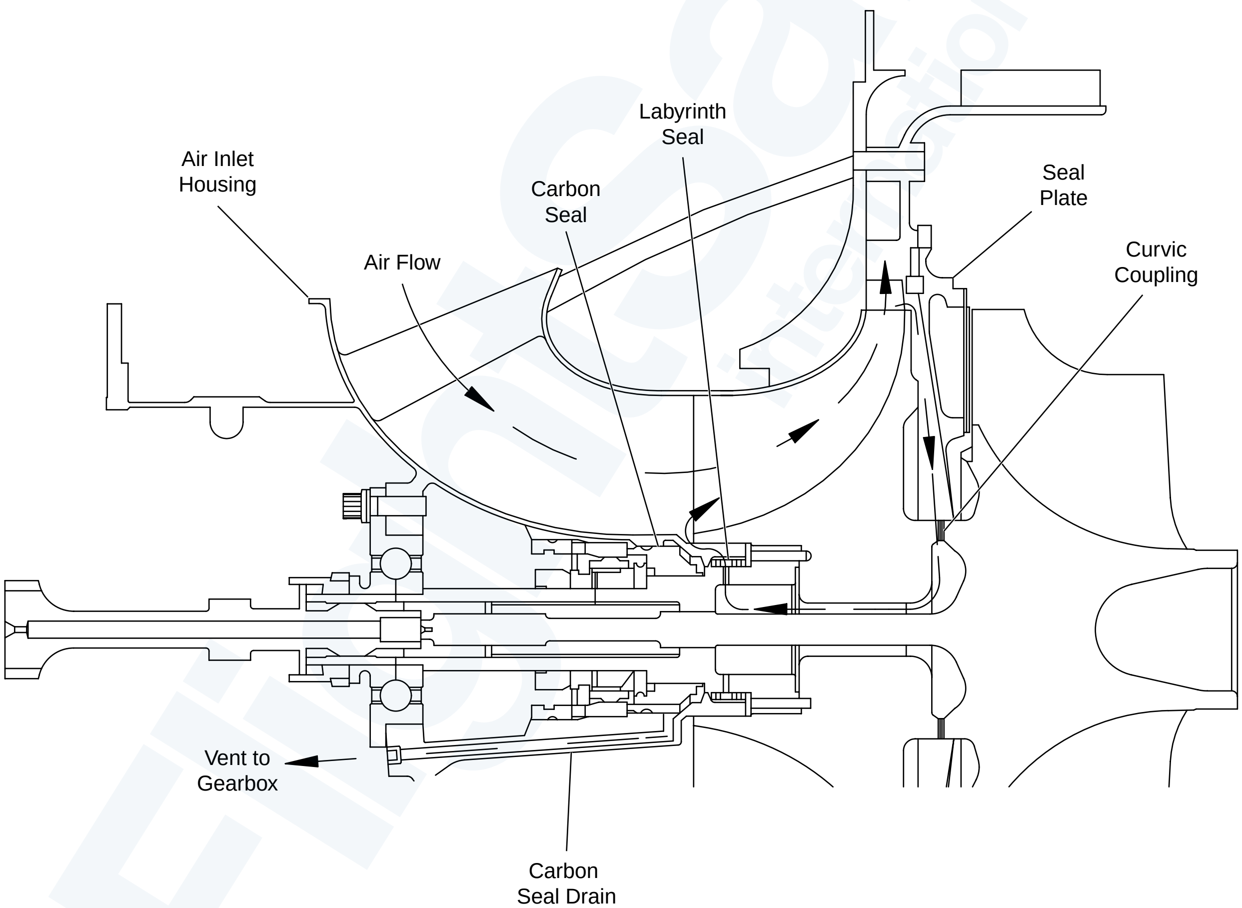


Figure 49-47. Carbon and Labyrinth Seals

Labyrinth Seal

The Labyrinth Seal is pressed on to a machined shoulder located on the hub of the compressor wheel. The Labyrinth Seal prevents oil from entering the compressor.

Gearbox Output Shaft Seal

Refer to Figure 49-48. Gearbox Output Shaft Seal.

The gearbox output shaft seal is installed at the generator mount pad. The carbon seal prevents oil in the bearing area from entering the generator.

NOTE

The pre-Hamilton Sundstrand SB4503067-49-17 lip seal and the post-Hamilton Sundstrand SB4503067-49-17 carbon seal are not interchangeable at the aircraft level.

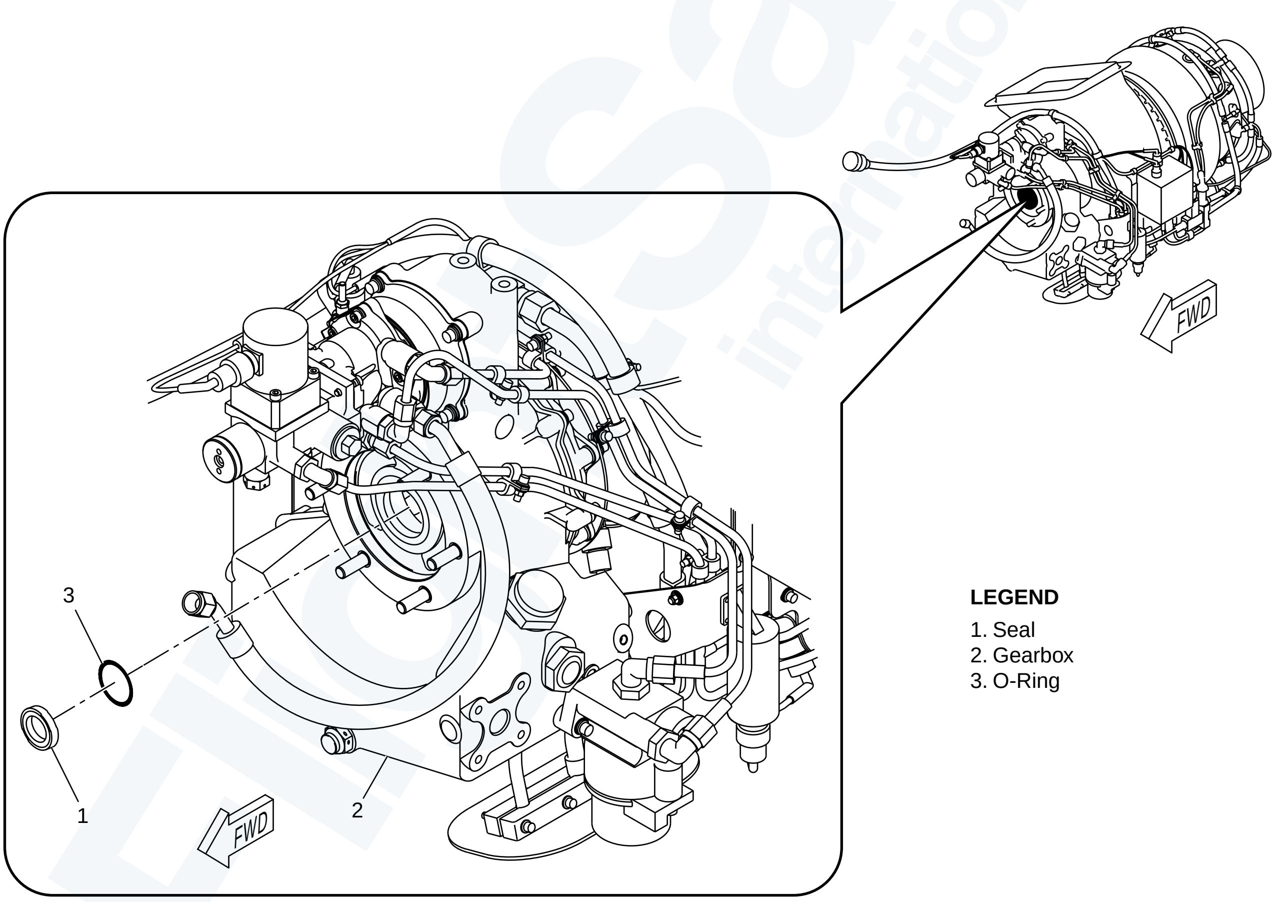


Figure 49-48. Gearbox Output Shaft Seal

Gearbox Vent Seal

Refer to Figure 49-49. Gearbox Vent Seal.

The APU gearbox vent seal is installed in the gearbox vent cap.

Ref: Task 49-90-13-000-801

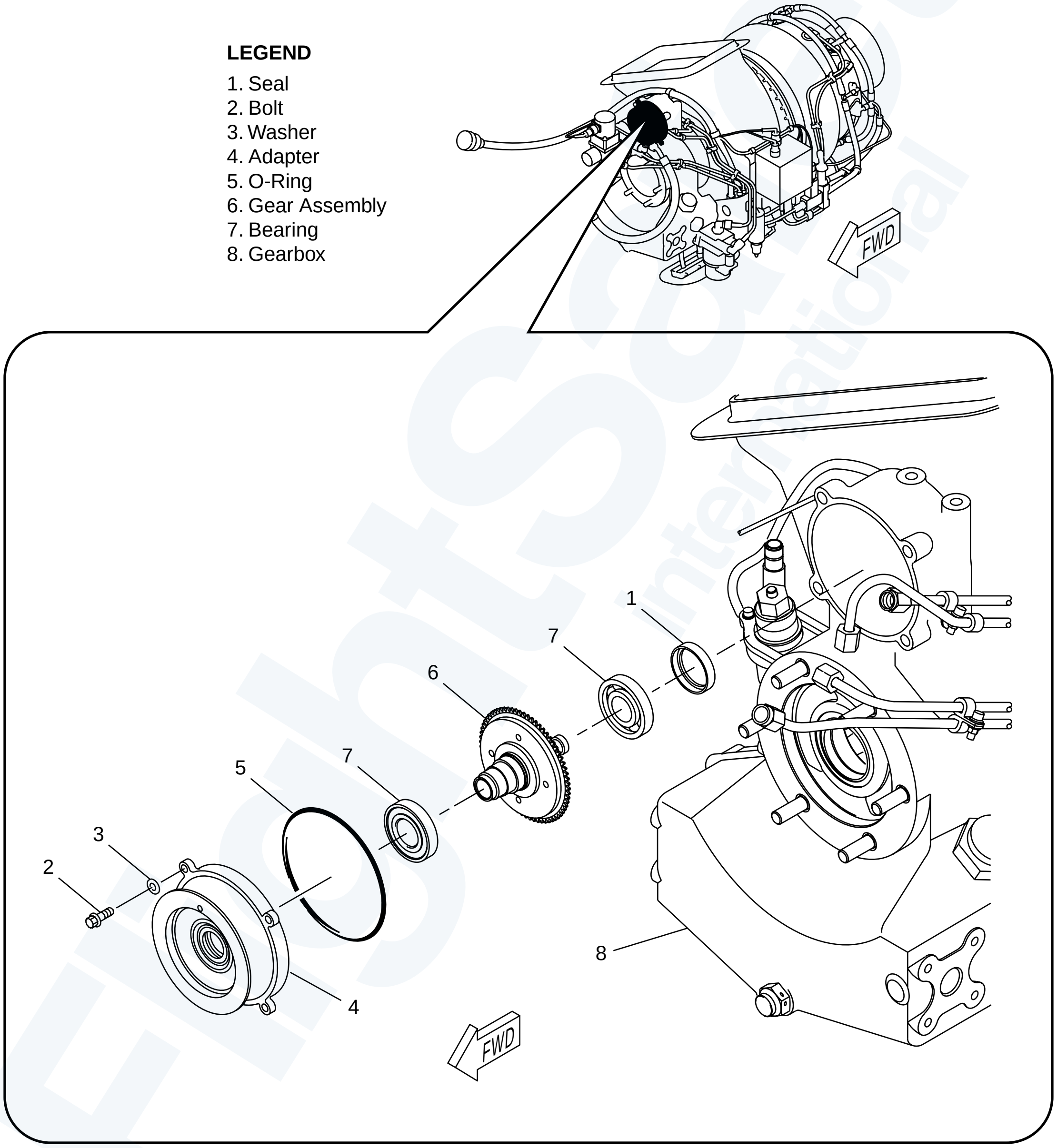


Figure 49-49. Gearbox Vent Seal

Oil Filler Port

Refer to Figure 49-50. Oil Filling, Draining and Checking.

The oil system is serviced through the oil filler port on the left side of the APU. The locking cap has a dip stick that is marked ADD and FULL.

Oil Sight Glass

The Oil Sight Glass is located below the oil filler port on the left side of the APU and supplies a visual indication of the oil level.

Magnetic Drain Plug

A magnetic drain plug is located on the lower front side of the gearbox housing. The drain plug has a magnetic chip detector that attracts ferrous metal particles in the oil.

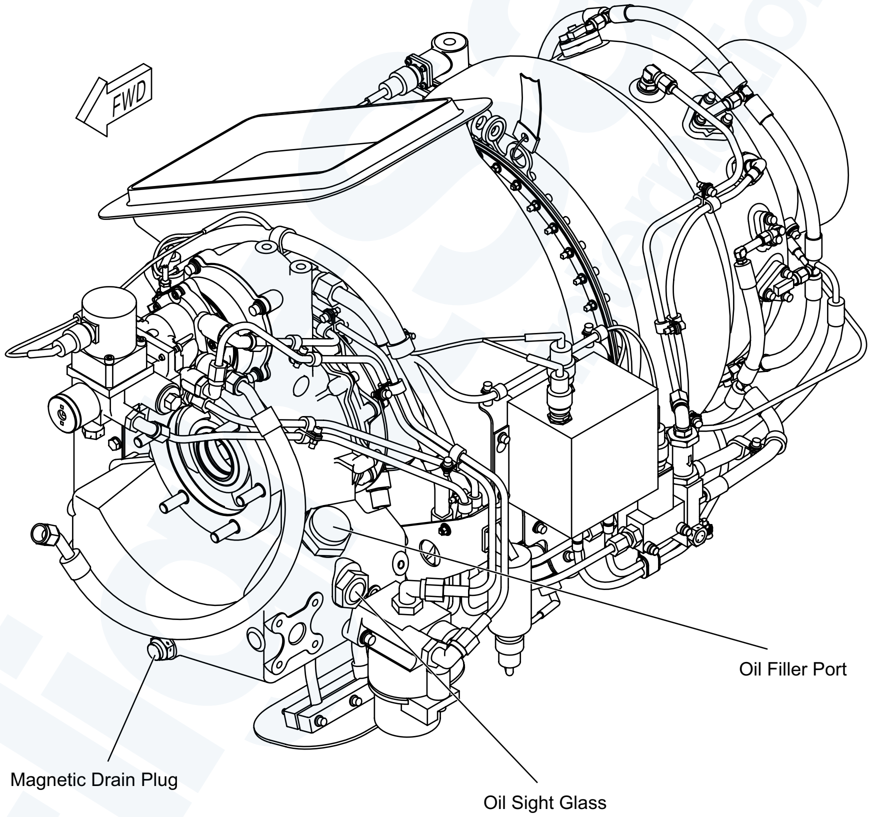


Figure 49-50. Oil Filling, Draining and Checking

OPERATION

Refer to Figure 49-44. Oil System Schematic.

The oil pump supplies oil flow through the fuel/oil cooler and filter, past the high oil temperature, low oil pressure switches to the gears, bearings and oil/air separator. The oil returns to the oil sump by gravity and the air is vented overboard through the drain mast vent. The oil pressure relief valve maintains the system oil pressure.

The fuel/oil cooler bypass valve opens if the cooler becomes plugged. If this happens there is no indication.

If the oil pressure indicator button is out, the oil filter is partially blocked. The oil filter bypass valve opens when the filter becomes more blocked.

The FADEC sends 28 VDC to energize the deprime solenoid valve during startup and shutdown to stop oil flow. Above 50% Ng the deprime valve is de-energized to allow the oil pump to produce oil flow.

The normally open high oil temperature switch closes when the oil is hot. The closed switch causes the FADEC to shut down the APU.

On start up of the APU, the normally closed Low Oil pressure switch opens when oil pressure is greater than 35 psi and will stay open during normal APU operation.

Oil pressure will not be monitored until the APU has reached 100% operating speed and the switch has been open for 12 seconds. When the oil pressure is less than 35 psi (241 kPa), the switch close supplying the closed loop signal from the FADEC, this will cause the FADEC to shut down the APU.

MAINTENANCE PRACTICES

The following are abbreviated descriptions of the maintenance practices and are intended for training purposes only. For a more detailed description of the practices, refer to the tasks in the Bombardier AMM PSM 1-84-2.

Oil Filter Element Replacement

NOTE

The APU oil filter element replacement cycle is every 3,000 hours of APU operation.

Remove oil filter retainer from the oil-filter bypass valve. Remove the oil-filter bypass Valve and discard the oil filter element.

Install the oil filter element with new O-ring using assembly fluid to lubricate the O-ring and seals. Install the bypass valve and retainer.

Oil Level Check of the APU

Clean the APU sight gauge and make sure the oil level is above the base of the gauge. If the oil level is below the base of the sight gauge, add the same type of approved oil to the yellow center mark on the sight gauge.

Servicing of the APU Oil

Remove the oil drain plug and drain the used oil. Carefully fill the oil tank to the level at the yellow center mark on the sight gauge with an approved oil of the same type.

Do not mix different types of oil. If the type of oil in use is not available, fill the APU with the new oil type. Operate the APU for 5 minutes then drain the oil from the APU tank.

Change the APU oil and filter again with the new oil type. Operate the APU for 5 minutes. Check the oil level of the APU again.

If the float ball is at the top of the sight gauge, remove the excess oil.

49-00-00 SPECIAL TOOL & TEST EQUIPMENT

- PF51-040-1 Hoist - APU
- 4503193 APU Build-Up Stand
- 400-76247-1 Lifting Lug
- GSB2400001 Digital Multimeter, Handheld
- AID-00536 Lip seal installation tool
- ST90889 Oil seal assembly bullet
- Commercially Available (32 MB RAM, 35 MB Hard Drive Space, RS-232 Serial Port)
- AGE70216 Cable-APU, FADEC to Laptop Interface Adapter (25 ft. cable RS485 to RS232).
- ERR 3186 “Smart Terminal” User Guide
- Hamilton Sundstrand Web Site http://www.hs-powersystems.com
- HS Power System Phone: 858–627–6770 “Smart Terminal” T46C12 Download (Laptop Trouble Shooters)
- 4503277, Software PN “Smart Terminal” T46C12 FADEC Software, (Contact Hamilton Sundstrand for current part number)
- ST 70109 Lip seal removal tool
- ST 90889 Lip seal installation tool
- AGE70051 Seal Puller (pre-Hamilton Sundstrand SB 4503067-49-17)
- AGE70660 or AID-01385 Seal Extraction Tool (post-Hamilton Sundstrand SB 4503067-49-17)
- AID 00606 Pusher (pre-Hamilton Sundstrand SB 4503067-49-17 only)
- AID-01357 Lip seal install tool or AGE70661 Seal insertion tool

49-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 49-00-00-869-801: Start Procedure of the APU.
- AMM 49-00-00-869-802: Shut Down Procedure of the APU.
- AMM 45-00-49-742-801: Retrieval of Data from the Central Diagnostic System (CDS) - Auxiliary Power Unit (APU).
- AMM 49-00-00-710-801: Operational Test of the Auxiliary Power Unit.
- AMM 49-00-00-720-801: Functional Test of the Auxiliary Power Unit.
- AMM 49-20-01-000-801: Removal of the Engine Assembly.
- AMM 49-20-01-400-801: Installation of the Engine Assembly.
- AMM 49-30-01-000-801: Removal of the Fuel Control Assembly.

- AMM 49-30-01-400-801: Installation of the Fuel Control Assembly.
- AMM 49-40-04-210-801: General Visual Inspection of the APU Igniter Plug.
- AMM 49-40-04-710-801: Operational Test of the APU Igniter Plugs.
- AMM 49-60-04-742-801: Retrieval of Data from the FADEC with the Smart Terminal.
- AMM 49-70-00-710-801: Operational Test of the APU Fault Code Indications (This task incorporates 494000-201 and 499000-202) (MRB #497000-201).
- AMM 49-90-10-000-801: Removal of the Oil Filter Element.
- AMM 49-90-10-400-801: Installation of the Oil Filter Element
- AMM 12-10-49-210-802: Oil Level Check of the APU.
- AMM 12-10-49-680-801: Draining of the APU Oil System.